

Generative Bridge Search: A Relation-First Method for Producing Candidate Bridge Solutions

Subtitle: Typed Blocker-Unblocker Matching, Declared Adaptation, and Verification Obligations **Author:** Artur Motruk **Affiliation:** Independent researcher **ORCID:** 0009-0003-6928-1701 **Programme:** Relation-First Organization Programme **Document role:** Late foundation / core generative methodology **Public release status:** Repository manuscript

Abstract

Bridge-Valence Diagnostics classifies bridge problems and prescribes actions. It tells us when a problem is bounded, finite-unifying, critical, mixed, underdeclared, or empirical. It also tells us what action follows: close, split, calibrate, map the atlas, or search for compatibility operators.

But diagnosis is not generation.

Central claim: BVD and Generative Bridge Search are reciprocal faces of a single RFOP bridge-problem state. BVD reads the **question-shape**; GBS reads the **candidate-answer-shape**. The mediation is:

$$M_{QA} = G_{\text{gen}} \circ \pi_{\text{Bl}} \circ Q_{\text{BVD}}.$$

Thus the analytical and generative layers are not independent methodologies. They are paired readouts of one programme-native bridge state.

This paper begins the next methodology layer:

How does the programme generate candidate solutions?

The answer proposed here is:

RFOP generates candidates by typed blocker-unblocker matching and declared adaptation.

Given a target blocker profile:

$$\mathfrak{Bl}_T,$$

the programme searches a relation atlas for resolved or partially resolved blocker profiles:

$$\mathfrak{Bl}_j$$

equipped with unblocker packets:

$$\mathfrak{U}_j.$$

If there exists a typed structural match:

$$\Theta : \mathfrak{Bl}_j \rightarrow \mathfrak{Bl}_T,$$

then the programme generates an adapted candidate unblocker:

$$\mathfrak{U}_T^* = \text{Adapt}_\Theta(\mathfrak{U}_j, T).$$

The output is not a verified solution. It is a candidate compatibility operator, construction, representation, duality, quotient, witness, or calibrator, together with explicit verification obligations.

Thus:

$$\boxed{\text{structural match} + \text{declared adaptation} \Rightarrow \text{candidate unblocker.}}$$

But:

$$\boxed{\text{candidate unblocker} \not\Rightarrow \text{verified solution.}}$$

Verification remains substantive mathematical, empirical, or field-specific work.

The method is governed by explicit guardrails: no analogy without a typed blocker match; no import without declared adaptation; no candidate without verification obligations; no novelty claim without trace; and no bypassing No-Free-Transfer. These guardrails prevent the generative procedure from becoming loose analogy, source-context theorem laundering, or unverified solution assertion.

Diagnostic tables from BVD and sustained bridge-valence case studies become seed sets for generative search. A row in a diagnostic table may function as a target blocker profile, a source blocker profile, or a candidate-unblocker search target. Candidate generation therefore proceeds through the trace:

$$\boxed{\text{target seed row} \rightarrow \text{blocker profile} \rightarrow \text{source candidate rows} \rightarrow \text{typed match} \rightarrow \text{adaptation output} \rightarrow \text{verification}}$$

The full mediation from question-shape to candidate-answer-shape is:

$$\pi_{\text{Bl}} : \mathfrak{Q}_T \rightarrow \mathfrak{Bl}_T$$

followed by the generative search operator:

$$G_{\text{gen}}(\mathfrak{Bl}_T, \mathfrak{A}_U, \mathcal{K}_T).$$

Thus:

$$\boxed{\text{question-shape} \rightarrow \text{blocker profile} \rightarrow \text{candidate-answer-shape.}}$$

Candidate status is a calibrated spectrum rather than an ontological shortcut:

rejected analogy < weak candidate < generated pending < high-priority candidate < admitted solution.

This is the coarse granularity $\Gamma_{\text{gen}}^{(5)}$. Admitted solution is not reached by priority score alone; it requires a separate verification/admission gate:

$$\text{Adm}_{\mathcal{K}_T}(\mathfrak{U}_T^*) = 1.$$

A declared decision vector records the generative score at a chosen granularity:

$$\boxed{\mathcal{G}_{\text{gen}} = (\theta_{\text{match}}, c_{\text{core}}, s_{\text{shell}}^-, \kappa_{\text{cand}}, v_{\text{obl}}, g_{\text{guard}}).$$

The paper’s contribution is therefore not an automatic solver. It is a programme-native candidate-generation layer with explicit source basis, typed match, adaptation, residual expectation, and verification obligation.

0. Source basis

Source	SHA256 prefix	Role
bridge_valence_diagnostics_predictive_criticality	2351212898f33c5	Bridge-Valence Diagnostics / action-space rulebook
rfop_bridge_valence_self_audit_v0_6_reproducibility	061956e4802ccea9	source-admitted RFOP self-audit / high-stakes branch map
bridge_valence_diagnostic_atlas_causal_set_theorem	67890c89571b0dd6	CSLTS sustained application / reproducibility packet

0.1 Programme dependency

This paper depends on the following programme machinery:

Programme machinery	Role here
Bridge-Valence Diagnostics	identifies blocker profiles and action spaces
Recursive Bridge Calibration	supplies calibrators and verification obligations
Recursive Chart Learning	supplies branch-field search and trace update
Triadic Closure / Substitution	supplies typed role substitution and structural matching
Closure Atlases	supply the search space of related blocker/unblocker forms
No Undeclared Relation	blocks undeclared imports
No-Free-Transfer	blocks illegal transfer of source-context solutions
Charted Organization	makes the generated candidate a charted readout, not an untyped analogy

0.2 Programme-machinery dependency spine

The generative layer is not a new free-standing heuristic. Its components are generated by existing RFOP machinery.

Generative component	Programme source	Role
blocker profile $\mathfrak{B}l_T$	BVD diagnostic state + theorem/no-go activation	extracts the typed obstruction to search on
unblocker packet $\mathfrak{U}_j$	RBC + source-admitted theorem/bridge papers	packages a solved/reduced blocker with scope and proof structure
unblocker atlas $\mathfrak{A}_U$	closure atlas + branch field	stores candidate source mechanisms by blocker type
typed match Θ	triadic substitution + charted organization	maps source blocker form to target blocker form under declared role substitution
transfer core/shell split	No-Free-Transfer + NUR	separates transferable structure from context-specific conditions
adaptation operator Adapt_Θ	RBC + triadic substitution	produces target-side candidate plus proof obligations
verification obligations Π_T^*, Obl_T^*	RBC calibrator and admission rule	prevents candidate generation from becoming solution assertion
traceable novelty	RCLT trace discipline	records source, match, adaptation, residuals, and update

Generative component	Programme source	Role
guardrails	NUR/NUB/NUO/NFT	block analogy, import, unverifiable novelty, and theorem laundering

Theorem 0.3 — Programme-Native Generation Theorem

Every formal component of Generative Bridge Search is obtained by one of five programme-native operations:

1. diagnostic readout;
2. closure-atlas search;
3. typed triadic substitution;
4. calibrated adaptation;
5. admitted trace/update.

Therefore Generative Bridge Search is a generative layer internal to RFOP, not an external analogy engine.

Proof

The blocker profile is a readout of a BVD diagnostic state. The unblocker atlas is a closure-atlas/branch-field structure. The typed match is a triadic substitution/charted-readout map. The adaptation operator is a calibrated bridge update governed by RBC. The verification obligations and admission conditions are imposed by RBC, NUR, and No-Free-Transfer. The generation trace is recorded by RCLT. Hence the whole procedure is generated by existing RFOP operations. $\square$

Corollary 0.4 — Generation extends compatibility search

BVD defines the critical-locus action:

$$\text{Act}_{\text{crit}} = (\text{AtlasMap}, \text{CompatSearch}).$$

Generative Bridge Search supplies the internal structure of:

$$\text{CompatSearch}.$$

It does not replace BVD. It extends the compatibility-search component of BVD.

0.5 Full-programme derivation spine

The generative paper must not be read as a new methodology invented after BVD. It is the next layer in the existing programme chain.

The derivation spine is:

relation-first $\rightarrow$ organization $\rightarrow$ charted visibility $\rightarrow$ learning trace $\rightarrow$ triadic closure $\rightarrow$ closure atlas $\rightarrow$ no-un

0.6 Programme machinery coverage table

Programme layer	What it contributes to generation
Relation-first primitive	every generated candidate is a relation claim, never an object-import

Programme layer	What it contributes to generation
Theory of Organization	candidate generation must preserve declared task-visible organization, not arbitrary resemblance
Charted Organization Theory	structural matches and generated candidates are charted readouts; no chartless information
Recursive Chart Learning Theory	generation is admitted path/branch update through a chart-visible branch field
Triadic Closure	blocker profiles decompose into boundary/mediation/readout faces
Triadic Substitution	typed structural match is role-substitution under declared preservation
Closure Atlases	unblocker search occurs in a closure atlas of branch/problem forms
No Undeclared Relation	no source-to-target transfer without declared bridge relation
No Undeclared Boundary	no candidate admissibility without boundary/tolerance/scope
No Undeclared Observer	no candidate readout without observer/evaluator/calibrator
No-Free-Transfer	source-context solution does not transfer for free
Recursive Bridge Calibration	adaptation produces proof obligations under a target calibrator
Bridge-Valence Diagnostics	supplies question-shape, blocker profile, action space, and diagnostic seed rows
Generative Bridge Search	supplies candidate-answer-shape, not admitted solution

Theorem 0.7 — Full-Programme Derivation Theorem

Generative Bridge Search is derivable from RFOP first principles as the composition of:

1. relation-first bridge declaration;
2. charted diagnostic readout;
3. recursive branch-field search;
4. triadic structural substitution;
5. closure-atlas navigation;
6. calibrated adaptation;
7. guardrail-gated admission;
8. trace/update recording.

Proof

A generated candidate is a proposed source-to-target relation, so relation-first and No Undeclared Relation require source, bridge, target, boundary, and readout. The target problem must be charted and diagnosed, which gives the BVD question-shape. The relevant obstruction is extracted as a blocker profile. Recursive Chart Learning supplies the branch-field and trace/update interpretation of candidate search. Triadic closure decomposes the blocker into typed faces, and triadic substitution supplies the typed match between blocker profiles. Closure atlases supply the search space of possible source unblockers. Recursive Bridge Calibration supplies the adaptation operator and target-side verification obligations. No-Free-Transfer, NUB, and NUO impose the guardrails. The generated candidate is then recorded as a traceable update, not as an admitted solution. Therefore the method is derivable from RFOP first principles. $\square$

Corollary 0.8 — The paper is not merely an analogy method

Because every candidate must pass through blocker-profile projection, typed match, adaptation, guardrails, and verification obligations, the method is not analogy by resemblance.

Corollary 0.9 — The paper is not merely BVD restated

BVD produces the question-shape and action need. Generative Bridge Search produces the candidate-answer-shape by searching for and adapting unblocker packets. Thus it extends BVD rather than repeating it.

0.10 Derivation/import register

This section makes the source status of each generative object explicit.

Object / rule	Status	Programme source	What is imported or derived	Non-transfer boundary
$\mathfrak{Q}_T$ question-shape	imported/assembled	BVD	diagnostic state, pre/post vectors, residual/status/action projects diagnostic state to search-relevant obstruction	does not generate answers
π_{BI} blocker projection	derived here	BVD + RBC	active residual, subtree, target-type jump, calibrator, need packages source mechanism with proof, scope, calibrator	projection is not solution
$\mathfrak{B}_{l_T}$ blocker profile	derived here	BVD theorem/no-go activation + RBC obstruction vector	stores source-admitted unblocker packets	only valid for declared target
$\mathfrak{U}_j$ unblocker packet	derived/import schema	RBC + source branch papers	typed morphism between blocker profiles	source scope does not transfer
$\mathfrak{A}_U$ unblocker atlas	derived schema	closure atlas + RCLT branch field	separates transferable structure from context conditions	must be source-admitted
Θ typed match	derived here	triadic substitution + charted organization	produces target candidate + obligations	surface analogy rejected
Core / Shell split	derived here	NFT + NUR	searches admitted source packets and adapts them	shell does not transfer for free
Adapt_Θ	derived here	RBC + triadic substitution	scores match/adaptation/guardrails	candidate not admitted solution
G_{gen}	derived here	closure-atlas search + RCLT update		search result still pending
$\mathfrak{R}_{\text{gen}}^\Gamma$	derived calibrator form / chosen granularity	BVD scoring + RBC calibrator		priority score is not final; Γ is chosen per audit
$\Gamma_{\text{gen}}^{(5)}$ status chart	derived spectrum / selected coarse chart	BVD granularity theory	status spectrum readout	exact coarseness is an audit choice; finer charts may refine

Object / rule	Status	Programme source	What is imported or derived	Non-transfer boundary
guardrails	imported/adapted	NUR/NUB/NUO/NFT + RBC	prevent analogy/import/verification failures	cannot be disabled
M_{QA}	derived here	BVD + GBS	maps question-shape to candidate-answer-shape	candidate-answer not admitted answer
verification matrix	derived here in §14B	RBC	records target obligations and admission gate	domain proof remains external

Definition 0.11 — Derivation status

Every component in this paper has one of four statuses:

internal derivation

named programme import

declared calibrator/refinement

domain verification outside the method.

No component may be used without one of these statuses.

Theorem 0.12 — No Unregistered Machinery Theorem

Every formal component used by Generative Bridge Search must appear in the derivation/import register or be explicitly introduced as a local definition with its proof obligation.

Proof

A formal component in RFOP is a relation claim. By No Undeclared Relation, every relation claim must declare its source, target, mediation, and readout. By Recursive Bridge Calibration, every methodological component must state its calibrator/admission role. Therefore an unregistered component would be an undeclared bridge in the method itself. Hence every component must either be registered as a programme import, derived locally, declared as a calibrator, or marked as external domain verification. $\square$

Corollary 0.13 — The spine is source-typed

The derivation spine is not only narrated. It is typed by component source and admission role.

0.14 Derivation-exhaustion rule

The word “declared” must not be used as a convenience marker for machinery that the programme can derive.

The rule is:

derive if programme-internal; declare only if target/application-specific.

Definition 0.15 — Programme-internal component

A component is **programme-internal** when it concerns one of:

1. relation-first source/bridge/target typing;
2. charted readout;
3. task/evaluator/calibrator shape;
4. trace/admission/update structure;
5. triadic face/substitution/coupling structure;
6. no-go or no-free-transfer discipline;
7. BVD diagnostic coordinates/action spaces;
8. generative candidate structure.

Programme-internal components must be derived or imported from named programme papers.

Definition 0.16 — Application-specific residue

A component is an **application-specific residue** when it depends on a target context not fixed by RFOP alone.

Examples:

1. which external problem is chosen;
2. which target regime is being tested;
3. which tolerance is scientifically acceptable;
4. which source theorem family is available in an external domain;
5. which empirical data/instrument/loss model is used;
6. which score-chart granularity is chosen for a given audit.

Application-specific residues may be declared, but the declaration must be explicit.

Theorem 0.17 — No Convenient Declaration Theorem

No formal component of Generative Bridge Search may be introduced by declaration if it is programme-internal.

Proof

A programme-internal component is part of the RFOP relation machinery: source/bridge/target typing, charted readout, trace/update, triadic substitution, calibrator shape, or no-go discipline. These are already governed by existing programme papers. If such a component were merely declared, the paper would introduce a hidden primitive or an unsupported method step. That would violate No Undeclared Relation at the level of the methodology itself. Therefore programme-internal components must be derived or imported from named programme papers. □

Corollary 0.18 — Declarations are allowed only at the target boundary

Declarations remain legitimate for target-specific choices because RFOP cannot derive an external target's tolerance, data, theorem availability, or empirical convention without importing that target domain.

Thus:

method shape is derived;

application parameters are declared.

0.19 Declaration-residue audit

Component	v0.9 status	v0.10 status	Reason
blocker profile	derived	derived	BVD residual + active subtree + RBC obstruction
unblocker packet schema	derived/import schema	derived	RBC bridge packet with source/scope/proof fields
source-unblocker population	application-declared	source-admitted external input	RFOP cannot derive external theorem inventory
typed match Θ	derived	derived	triadic substitution over blocker graph
match admissibility	partly declared	derived criterion + target tolerance residue	preservation rules derived; tolerance target-specific
adaptation operator	derived	derived	RBC update + triadic substitution + NFT shell split
core/shell split	derived	derived	No-Free-Transfer forces it
generative score sheet	declared calibrator	general form derived; granularity declared	BVD derives calibrator form; granularity target-specific
$\Gamma_{\text{gen}}^{(5)}$	declared coarse chart	declared residue over derived spectrum	exact coarseness is audit choice
guardrails	imported/derived	derived/imported	NUR/NUB/NUO/NFT + RBC
verification matrix	derived	derived	RBC admission + NUR/NUB/NUO
matrix entries	declared	application-specific residue	target problem supplies hypotheses/proofs/data
candidate-answer-shape	derived	derived	BVD question-shape + blocker projection + GBS

Theorem 0.20 — Declaration Residue Theorem

After applying the derivation-exhaustion rule, the only legitimate declarations remaining in Generative Bridge Search are target/application parameters and granularity choices.

Proof

The method-internal components are blocker profiles, typed matches, unblocker packet schema, adaptation, guardrails, score-sheet form, candidate status spectrum, and verification matrix shape. Each is derived or imported from RFOP programme machinery: BVD, RCLT, RBC, triadic substitution, closure atlases, and no-go discipline. The remaining choices concern external target regime, tolerance, source theorem availability, empirical data/loss, and score-chart granularity. These are not derivable from RFOP alone without overclaiming control over the target domain. Therefore only target/application parameters and granularity choices remain legitimately declared. $\square$

1. Problem

The Bridge-Valence Diagnostic gives:

$$\text{Diag}(X, D) = (C, \Delta_{\text{diag}}, m_C, \text{Status}).$$

It can say:

this is a critical locus

and then prescribe:

AtlasMap + CompatSearch .

But the open methodological question is:

How does CompatSearch generate candidates?

Without a generation theory, the method risks stopping at organized decomposition:

diagnose $\rightarrow$ decompose $\rightarrow$ hand off to researchers.

That is valuable, but it is not yet generative.

A generative method must produce candidate solution mechanisms:

candidate compatibility operators, witnesses, dualities, representations, quotients, calibrators, or construction.

2. From analogy to typed structural match

This paper rejects loose analogy as a generative principle.

A loose analogy has the form:

problem A feels similar to problem B .

That is not RFOP-admissible.

The programme-native form is:

the blocker profile of A matches the blocker profile of B under declared typed substitution.

Thus, generation begins not with resemblance between fields, but with a typed relation between obstruction structures.

2A. Relation to structure-mapping and analogical reasoning

The method is adjacent to structure-mapping theories of analogy associated with Gentner, Holyoak, and related cognitive-science accounts of productive analogical transfer.

The shared point is:

productive transfer depends on relational structure, not surface vocabulary.

However, Generative Bridge Search is stricter in four ways.

Structure-mapping / analogical reasoning	Generative Bridge Search
maps relational structure between source and target domains	maps typed blocker graphs between diagnosed bridge problems
distinguishes surface similarity from structural alignment	rejects surface analogy unless a typed blocker-match morphism exists
can suggest candidate inferences	generates candidate unblockers with verification matrices
evaluates analogy by coherence, systematicity, and constraints	evaluates candidate generation by BVD/RBC score, guardrails, and no-free-transfer discipline

Thus the paper does not deny the relevance of analogy research. It gives a programme-native formalization of a stricter object:

typed blocker-unblocker transfer under declared verification obligations.

The key difference is admission discipline. Structure-mapping can explain why a mapping is cognitively productive. GBS asks whether the mapping is admissible as a candidate bridge under RFOP no-go, calibration, and verification rules.

Proposition 2A.1 — Typed match refines structure-mapping

A typed blocker-match morphism is a structure mapping equipped with programme-native type, boundary, readout, transfer, and verification constraints.

Proof

A structure mapping preserves relational organization between a source and target. A typed blocker-match morphism preserves relational organization between blocker graphs while additionally preserving or explicitly retyping face labels, target-type jumps, unblocker need, calibrator nodes, and dependency edges. It also records shell burden and verification obligations. Therefore typed blocker matching refines ordinary relational structure mapping by adding RFOP type and admission constraints. $\square$

3. Blocker profiles

Definition 3.1 — Bridge problem state

A bridge problem state is:

$$\mathfrak{P}_T = (X_T, D_T, \mathfrak{B}_T, \mathcal{K}_T, \mathfrak{D}_T),$$

where:

- X_T is the source presentation;
- D_T is the target distinction family;

- $\mathfrak{B}_T$ is the current bridge attempt or bridge class;
- $\mathcal{K}_T$ is the target calibrator;
- $\mathfrak{D}_T$ is the BVD diagnostic state.

Definition 3.2 — Blocker profile

The **blocker profile** of a bridge problem state is:

$$\mathfrak{Bl}_T = (\Delta_T, \mathcal{A}_T, \tau_T, \chi_T, \mathcal{K}_T, \text{Need}_T).$$

Where:

Component	Meaning
Δ_T	active residual or obstruction family
$\mathcal{A}_T$	active theorem/no-go subtree
τ_T	blocked target-type transfer
χ_T	face / triad / substitution structure of the blocker
$\mathcal{K}_T$	calibrator under which resolution would be admitted
Need_T	required unblocker role

The blocker profile is the search key.

It is not the field name.

For example:

Hauptvermutung

is not the blocker profile. The blocker profile may be:

one discrete presentation admits multiple continuum targets; no declared similarity operator collapses them.

Definition 3.3 — Unblocker need

The unblocker need is the role a candidate must play to downgrade or resolve the blocker.

Examples:

$\text{Need}_T \in \{\text{compatibility operator, representation theorem, duality, witness construction, quotient criterion, gluing}\}$

4. Unblocker packets

Definition 4.1 — Unblocker packet

An **unblocker packet** is:

$$\mathfrak{U}_j = (P_j, \mathfrak{Bl}_j, \kappa_j, \mathcal{K}_j, \Pi_j, \text{Scope}_j).$$

Where:

Component	Meaning
P_j	source problem/context where the mechanism works
$\mathfrak{Bl}_j$	blocker profile resolved or partially resolved in that context
κ_j	compatibility operator / construction / theorem / mechanism
$\mathcal{K}_j$	calibrator under which it is admitted
Π_j	proof or verification structure
Scope_j	declared scope and non-transfer boundaries

An unblocker packet is not just “a solution.” It is a solution mechanism with its own declared boundary, target, proof, and non-transfer conditions.

Definition 4.2 — Unblocker atlas

A **structural unblocker atlas** is a collection:

$$\mathfrak{A}_U = \{\mathfrak{U}_j\}_{j \in J}$$

indexed not by discipline but by blocker-profile type.

Candidate families include:

duality theorems, representation theorems, spectral constructions, quotient criteria, gluing theorems,

5. Typed structural match

Definition 5.1 — Typed structural match

A **typed structural match** from a source blocker profile to a target blocker profile is a declared map:

$$\Theta : \mathfrak{Bl}_j \rightarrow \mathfrak{Bl}_T$$

that preserves, to declared tolerance:

1. active face structure;
2. target-type transfer shape;
3. no-free-transfer obstruction type;
4. required unblocker role;
5. calibrator role;
6. proof-obligation shape.

A match is not required to preserve source vocabulary.

Indeed, generative matches often do **not** preserve vocabulary.

They preserve obstruction form.

Definition 5.2 — Match admissibility

A match Θ is admissible only if:

$$\text{Adm}_{\text{match}}(\Theta; \mathcal{K}_T) = 1.$$

This requires:

1. target-side declarations are complete enough to compare;
2. source-side unblocker scope is known;
3. no-free-transfer violations are explicitly identified;
4. the match preserves the relevant blocker structure;
5. a verification path is declared.

Definition 5.3 — False analogy

A proposed match is a **false analogy** when it preserves surface vocabulary but not blocker structure. Formally:

$$\Theta_{\text{vocab}} \text{ exists but } \Theta_{\text{block}} \text{ fails.}$$

False analogies are blocked by No Undeclared Relation and No-Free-Transfer.

5.4 Typed blocker graph

To make typed structural match reproducible, represent a blocker profile as a finite typed graph:

$$\mathcal{G}(\mathfrak{Bl}) = (V_{\text{face}}, V_{\text{target}}, V_{\text{transfer}}, V_{\text{need}}, V_{\text{cal}}, E_{\text{dep}}, \ell).$$

Where:

- V_{face} are active triadic faces;
- V_{target} are target-presentation nodes;
- V_{transfer} are target-type jumps / NFT sites;
- V_{need} are required unblocker roles;
- V_{cal} are calibrator/readout nodes;
- E_{dep} are dependency edges;
- ℓ is a type-labeling function.

Definition 5.5 — Typed-match morphism

A typed-match morphism is a partial structure-preserving map:

$$\Theta : \mathcal{G}(\mathfrak{Bl}_j) \rightarrow \mathcal{G}(\mathfrak{Bl}_T)$$

such that:

1. face-type labels are preserved or explicitly retyped by declared triadic substitution;
2. target-type jumps map to target-type jumps;
3. unblocker need maps to compatible unblocker need;
4. calibrator nodes map to calibrator nodes;
5. dependency edges preserve source/target incidence where mapped;
6. unmapped source-shell nodes are recorded as shell replacement burden.

Definition 5.6 — Match domain and shell complement

The **transfer core** of Θ is:

$$\text{Dom}_{\text{core}}(\Theta) \subseteq \mathcal{G}(\mathfrak{Bl}_j),$$

the subgraph on which structure is preserved.

The **shell complement** is:

$$\text{Shell}^-(\Theta) = \mathcal{G}(\mathfrak{Bl}_j) \setminus \text{Dom}_{\text{core}}(\Theta),$$

together with any target-side nodes required but not supplied by the source.

This graph-theoretic split is the formal version of the transfer-core / context-shell split.

Theorem 5.7 — Typed Match Is Triadic Substitution on Blocker Graphs

Every admitted typed-match morphism is an instance of triadic substitution and charted readout applied to blocker profiles.

Proof

A blocker graph records active triadic faces, target roles, target-type jumps, calibrators, and dependencies. A typed-match morphism maps a source role structure into a target role structure while preserving or explicitly retyping faces, jumps, needs, and calibrators. This is exactly a triadic role-substitution with declared landing conditions. Since the map is charted by ℓ and dependency edges, it is also a charted readout rather than an untyped analogy. $\square$

Corollary 5.8 — Match disagreement is local

A reader disputing a match must identify a failed preservation condition: face type, target jump, need, calibrator, dependency edge, or shell burden.

This makes match evaluation replicable.

6. Adaptation

Definition 6.1 — Adaptation operator

Given an admissible typed match:

$$\Theta : \mathfrak{Bl}_j \rightarrow \mathfrak{Bl}_T$$

and an unblocker packet:

$$\mathfrak{U}_j,$$

the **adaptation operator** produces:

$$\text{Adapt}_{\Theta}(\mathfrak{U}_j, T) = (\kappa_T^*, \mathcal{K}_T^*, \Pi_T^*, \Delta_T^*, \text{Obl}_T^*).$$

Where:

Component	Meaning
κ_T^*	candidate target-side compatibility operator / construction
$\mathcal{K}_T^*$	target-side calibrator refinement
Π_T^*	verification proof obligations
Δ_T^*	residuals expected to remain after adaptation
Obl_T^*	explicit tasks needed to verify or reject the candidate

Definition 6.2 — Transfer core and context shell

An unblocker packet decomposes into:

$$\mathfrak{U}_j = (\text{Core}_j, \text{Shell}_j).$$

Where:

- Core_j is the structure-dependent mechanism;
- Shell_j is the source-context-specific condition set.

Adaptation transfers only what is preserved by Θ :

$$\Theta(\text{Core}_j) \rightsquigarrow \text{Core}_T^*.$$

The shell must be redeclared for the target context:

$$\text{Shell}_j \not\Rightarrow \text{Shell}_T.$$

This is the no-free-transfer guardrail.

Definition 6.3 — Adaptation distance

The **adaptation distance** is a qualitative/ordinal measure:

$$d_{\text{adapt}}(\mathfrak{U}_j, T)$$

recording how much of the source shell must be replaced.

Four regimes:

Regime	Meaning
direct import	source shell and target shell nearly identical
adapted transfer	core preserved, shell partly replaced
generative reformulation	only high-level blocker form preserved
false analogy	blocker form not preserved

7. Candidate generation

Definition 7.1 — Generative search operator

The **generative bridge search operator** is:

$$G_{\text{gen}}(\mathfrak{Bl}_T, \mathfrak{U}_U, \mathcal{K}_T) = \{\text{Adapt}_{\Theta_j}(\mathfrak{U}_j, T) : \Theta_j : \mathfrak{Bl}_j \rightarrow \mathfrak{Bl}_T \text{ admitted}\}.$$

The output is a set of candidate adapted unblockers.

Theorem 7.2 — Generative Bridge Search Theorem

Let T be a bridge problem with blocker profile $\mathfrak{Bl}_T$. Let $\mathfrak{U}_U$ be an unblocker atlas. If there exists an unblocker packet $\mathfrak{U}_j \in \mathfrak{U}_U$ and an admissible typed structural match:

$$\Theta : \mathfrak{Bl}_j \rightarrow \mathfrak{Bl}_T,$$

then the programme generates a candidate adapted unblocker:

$$\mathfrak{U}_T^* = \text{Adapt}_{\Theta}(\mathfrak{U}_j, T).$$

Proof

The typed match identifies a source blocker profile whose obstruction structure is preserved under declared substitution into the target blocker profile. The unblocker packet supplies a mechanism that resolves or reduces that blocker profile in its source context. The adaptation operator transports the structure-dependent core while replacing source-context shell conditions with target-context declarations. The output is therefore a target-side candidate unblocker plus verification obligations. This is exactly candidate generation. $\square$

Corollary 7.3 — Candidate generation is not solution verification

$$\mathfrak{U}_T^* \text{ generated} \not\Rightarrow \mathfrak{U}_T^* \text{ verified.}$$

A generated candidate becomes a solution only after it satisfies the target calibrator:

$$\text{Adm}_{\mathcal{K}_T}(\mathfrak{U}_T^*) = 1.$$

8. No-free-generation

Theorem 8.1 — No-Free-Generation Theorem

No candidate generated by structural unblocker transfer is admitted as a solution unless its target-side bridge, calibrator, and verification obligations are declared and satisfied.

Proof

A candidate generated from another context is a transfer claim. By No Undeclared Relation, the target relation must be declared. By No-Free-Transfer, the source-context solution mechanism cannot be promoted to the target context without a bridge law. By Recursive Bridge Calibration, the candidate must satisfy the target calibrator. Therefore candidate generation alone cannot admit a solution. $\square$

Corollary 8.2 — The method cannot bypass mathematics

The generative procedure does not replace proof, construction, experiment, or field-specific verification. It produces proof obligations.

Corollary 8.3 — The method blocks solution laundering

A source-context theorem cannot be laundered into a target-context theorem through vocabulary substitution.

Only structure-preserving, calibrator-declared adaptation is admissible.

9. Traceable novelty

Definition 9.1 — Traceable novelty

A generated candidate is **traceably novel** when:

1. its structural ancestry is recorded by Θ ;
2. its target-side formulation differs from the source mechanism due to adaptation;
3. it has target-specific verification obligations;
4. it is not merely a source theorem restated in new vocabulary.

9.2 Novelty regimes

Regime	Diagnostic form	Novelty status
direct import	Θ nearly identity; shell preserved	low novelty
adapted transfer	core preserved; shell partly replaced	moderate novelty
generative reformulation	high-level blocker form preserved; shell mostly replaced	high traceable novelty
false analogy	blocker form not preserved	inadmissible

Corollary 9.3 — Novelty is not magic

RFOP generation does not require pretending candidates arise from nowhere.

It records genealogy:

source unblocker $\rightarrow$ typed match $\rightarrow$ adapted candidate $\rightarrow$ verification.

10. Integration with critical-locus action space

Bridge-Valence Diagnostics defines:

$$\text{Act}_{\text{crit}} = (\text{AtlasMap}, \text{CompatSearch}).$$

This paper refines:

CompatSearch .

Specifically:

$$\text{CompatSearch} = \text{BlockerProfile} + \text{AtlasSearch} + \text{UnblockerMatch} + \text{Adapt} + \text{Verify}.$$

Thus generation is the internal structure of compatibility-operator search.

10A. BVD-derived generative score sheet

v0.4 made the worked examples end-to-end. But end-to-end is not the same as fully reproducible.

A reader still needed to ask:

why was this source row selected rather than another?

why is this typed match strong enough?

why is this generated candidate pending rather than rejected?

This section answers those questions by deriving a generative score sheet from the BVD score-sheet machinery.

10A.1 Generative diagnostic calibrator

A generative decision inherits the calibrator form from BVD/RBC and instantiates it for candidate generation:

$$\mathfrak{R}_{\text{gen}}^{\Gamma} = (D_{\text{gen}}, V_{\text{gen}}^{\Gamma}, \preceq_{\text{gen}}^{\Gamma}, \mathcal{E}_{\text{gen}}, \text{Adm}_{\text{gen}}, \text{Update}_{\text{gen}}).$$

Where:

- D_{gen} is the target: decide whether a candidate source row generates a target-side candidate unblocker;
- V_{gen}^{Γ} is the declared score domain;
- $\preceq_{\text{gen}}^{\Gamma}$ orders candidate strength / obstruction;
- $\mathcal{E}_{\text{gen}}$ records generative error;
- Adm_{gen} admits, rejects, or marks a candidate pending;
- $\text{Update}_{\text{gen}}$ updates the candidate after verification or failure.

Thus the existence and shape of the calibrator are derived; only the chosen granularity Γ is an application-level declaration.

10A.2 Generative decision vector

Define:

$$\mathcal{G}_{\text{gen}} = (\theta_{\text{match}}, c_{\text{core}}, s_{\text{shell}}^-, \kappa_{\text{cand}}, v_{\text{obl}}, g_{\text{guard}}).$$

Where:

Coordinate	Meaning	Positive or obstruction?
θ_{match}	typed blocker-match strength	positive
c_{core}	transfer-core preservation	positive
s_{shell}^-	source-shell replacement burden	obstruction
κ_{cand}	candidate compatibility coverage	positive
v_{obl}	verification-obligation clarity	positive
g_{guard}	guardrail satisfaction	required gate

10A.3 Count-style readouts

Where possible, use the following count-style readouts.

Generative coordinate	Count readout	L	M	H	VH
θ_{match}	number of preserved blocker fields among face, transfer, NFT, need, calibrator, proof-shape	0-1	2-3	4-5	6
c_{core}	number of source mechanism core elements preserved	0-1	2	3-4	≥ 5
s_{shell}^-	number/severity of source-shell assumptions needing replacement	0	1	2-3	≥ 4
κ_{cand}	target-face coverage of candidate	none	narrow	partial multi-face	broad/near-terminal
v_{obl}	verification obligations declared: target, assumptions, proof, residual, failure condition, calibrator	0-1	2-3	4-5	6
g_{guard}	guardrails passed out of five	0-2	3	4	5

For s_{shell}^- , higher is worse.

For g_{guard} , admissible candidate generation requires:

$$g_{\text{guard}} = VH$$

meaning all five guardrails pass.

10A.4 Candidate status rule

At the default coarse granularity $\Gamma_{\text{gen}}^{(5)}$, a candidate receives one of five statuses.

Status	Rule
rejected analogy	$\theta_{\text{match}} \leq M$ or $g_{\text{guard}} < VH$
weak candidate	$\theta_{\text{match}} = H$, $s_{\text{shell}}^- = H/VH$, and obligations are declared
generated pending	$\theta_{\text{match}} \geq H$, $c_{\text{core}} \geq H$, $g_{\text{guard}} = VH$, $v_{\text{obl}} \geq H$
high-priority candidate	generated pending plus $\kappa_{\text{cand}} \geq H$ and $s_{\text{shell}}^- \leq M$
admitted solution	generated candidate passes target calibrator $\text{Adm}_{\mathcal{K}_T} = 1$

10A.5 Priority score

For ranking only, define:

$$P_{\text{gen}} = \theta_{\text{match}} + c_{\text{core}} + \kappa_{\text{cand}} + v_{\text{obl}} + g_{\text{guard}} - s_{\text{shell}}^-.$$

This is not a truth score.

It is a search-priority score.

A candidate with high P_{gen} is worth verifying first. It is not thereby verified.

Theorem 10A.6 — BVD-Derived Generative Score-Sheet Theorem

The generative score sheet is programme-native because it is a granularity-indexed calibrator over BVD diagnostic rows and RFOP generative operations.

Proof

BVD supplies the general score-sheet form: a declared granularity, ordered readout values, evaluator, admission rule, and update rule. Generative Bridge Search applies this same form to a new target: deciding whether a source unblocker packet may generate a target-side candidate. The coordinates $\theta_{\text{match}}, c_{\text{core}}, s_{\text{shell}}^-, \kappa_{\text{cand}}, v_{\text{obl}}, g_{\text{guard}}$ are exactly the readouts required by typed structural match, adaptation, verification, and guardrail admission. Therefore the score sheet is not arbitrary. It is a BVD-derived calibrator for candidate generation. $\square$

Corollary 10A.7 — Generative decisions are challengeable row-by-row

A reader may challenge a generated candidate only by changing one of:

$$\theta_{\text{match}}, c_{\text{core}}, s_{\text{shell}}^-, \kappa_{\text{cand}}, v_{\text{obl}}, g_{\text{guard}},$$

or by changing the target/source row itself.

This localizes disagreement and blocks vibes-based candidate selection.

10B. Generative status spectrum and granularity refinement

The five candidate statuses are not primitive kinds.

They are a coarse chart over a finer generative viability spectrum.

The safe statement is:

candidate status = granularity-indexed readout of generative viability and admission state.

Definition 10B.1 — Generative viability presentation

For a generated candidate:

$$\mathfrak{U}_T^*,$$

define its **generative viability presentation**:

$$\mathfrak{V}_{\text{gen}}(\mathfrak{U}_T^*) = (\theta_{\text{match}}, c_{\text{core}}, s_{\text{shell}}^-, \kappa_{\text{cand}}, v_{\text{obl}}, g_{\text{guard}}, a_{\text{ver}}).$$

The first six coordinates are the generative decision vector. The final coordinate:

$$a_{\text{ver}}$$

records target-side verification/admission.

Thus:

$$a_{\text{ver}} = 0$$

means not verified;

$$a_{\text{ver}} = 1$$

means admitted by the target calibrator.

Definition 10B.2 — Coarse five-band status chart

The default status chart is:

$$\Gamma_{\text{gen}}^{(5)} = \{R < W < G < P < A\}.$$

Where:

Symbol	Status	Meaning
R	rejected analogy	match or guardrails fail
W	weak candidate	structurally suggestive but high burden or low coverage
G	generated pending	candidate produced with obligations but not prioritized
P	high-priority candidate	strong candidate worth verifying first
A	admitted solution	verification/admission gate passed

The order is a maturity/admission order:

$$R < W < G < P < A.$$

Definition 10B.3 — Admission gate

The transition:

$$P \rightarrow A$$

cannot be made by priority score alone.

It requires:

$$\boxed{\text{Adm}_{\mathcal{K}_T}(\mathfrak{U}_T^*) = 1.}$$

Therefore:

$$\boxed{P_{\text{gen}} \text{ high} \not\Rightarrow A.}$$

A high priority candidate is still only a candidate.

Definition 10B.4 — Finer generative granularity

A finer generative granularity:

$$\Gamma'_{\text{gen}}$$

refines $\Gamma_{\text{gen}}^{(5)}$ when there is a monotone collapse map:

$$q_{\Gamma' \rightarrow \Gamma^{(5)}} : V_{\text{gen}}^{\Gamma'} \rightarrow V_{\text{gen}}^{\Gamma^{(5)}}$$

such that the coarse status is recovered by:

$$S_{\Gamma^{(5)}} = q_{\Gamma' \rightarrow \Gamma^{(5)}} \circ S_{\Gamma'}.$$

For example:

$$W \rightsquigarrow (W_1, W_2, W_3),$$

$$G \rightsquigarrow (G_{\text{shell}}, G_{\text{proof}}, G_{\text{data}}, G_{\text{mixed}}),$$

$$P \rightsquigarrow (P_{\text{low}}, P_{\text{mid}}, P_{\text{high}}).$$

Definition 10B.5 — Boundary-sensitive candidate

A candidate is **boundary-sensitive** when small changes to:

$$\theta_{\text{match}}, c_{\text{core}}, s_{\text{shell}}^-, \kappa_{\text{cand}}, v_{\text{obl}}, g_{\text{guard}}$$

change its coarse status.

Such a candidate should be marked:

granularity-sensitive.

Theorem 10B.6 — Generative Status Spectrum Theorem

The candidate-status set:

{rejected analogy, weak candidate, generated pending, high-priority candidate, admitted solution}

is a coarse chart of a generative viability spectrum, not a primitive five-kind ontology.

Proof

The candidate status is determined by the generative decision vector, guardrail satisfaction, adaptation burden, compatibility coverage, verification obligations, and target admission. These are ordered or partially ordered readouts, not discrete primitive kinds. A coarse five-band chart partitions this ordered viability presentation into practical status bands. Since finer monotone refinements can split any band while preserving the coarse collapse, the five statuses are a coarse chart of a spectrum. $\square$

Corollary 10B.7 — Finer scoring is admissible

A future audit may refine the generative score sheet to:

$$\Gamma_{\text{gen}}^{(7)}, \quad \Gamma_{\text{gen}}^{(10)}, \quad \text{or a weighted/lexicographic chart.}$$

Such a refinement is admissible if it is declared, monotone, and preserves the admission-gate distinction.

Corollary 10B.8 — Admission is not merely the top score

The admitted-solution status is different from the highest priority status. It requires verification under the target calibrator.

Thus:

high-priority candidate $\neq$ admitted solution.

Register note

The generative statuses are like the BVD classifications: coarse enough to be usable, but refineable when the branch requires higher resolution.

10C. Question-shape, candidate-answer-shape, and mediation

The relation between BVD and Generative Bridge Search can now be stated directly.

BVD gives the question-shape.

Generative Bridge Search gives the candidate-answer-shape.

But this statement needs precision.

BVD does not give a natural-language question. It gives the formal shape of what is being asked by a target-bridge problem.

Generative Bridge Search does not give a final answer. It gives a candidate answer-shape: a structured family of possible unblockers with verification obligations.

Definition 10C.1 — Question-shape

For a target bridge problem T , the **question-shape** is:

$$\mathfrak{Q}_T = (X_T, D_T, \mathcal{D}_{\text{pre}}, \mathcal{D}_{\text{post}}, \Delta_{\text{diag}}, \mathcal{A}_T, \mathcal{B}_T, \text{Status}_T, \text{Act}_T).$$

Where:

- X_T is the source presentation;
- D_T is the declared target regime;
- $\mathcal{D}_{\text{pre}}$ is the BVD pre-diagnostic vector;
- $\mathcal{D}_{\text{post}}$ is the residual persistence / bridge-generation / audit-survival vector;
- Δ_{diag} is the diagnostic residual;
- $\mathcal{A}_T$ is the active theorem/no-go subtree;
- $\mathcal{B}_T$ is the current branch field;
- Status_T is the diagnostic status;
- Act_T is the prescribed action space.

This is the formal structure of the question.

Definition 10C.2 — Candidate-answer-shape

A **candidate-answer-shape** is:

$$\mathfrak{A}_T^* = (\text{Need}_T, \{\mathfrak{U}_{T,k}^*\}_{k \in K}, \{\text{Obl}_{T,k}^*\}_{k \in K}, \{\Delta_{T,k}^*\}_{k \in K}, \mathcal{K}_T^*, \text{Status}_{\text{gen}}).$$

Where:

- Need_T is the required unblocker role;
- $\mathfrak{U}_{T,k}^*$ are generated candidate unblockers;
- $\text{Obl}_{T,k}^*$ are verification obligations;
- $\Delta_{T,k}^*$ are residuals expected or left by each candidate;
- $\mathcal{K}_T^*$ is the target-side calibrator refinement;
- $\text{Status}_{\text{gen}}$ is the generative status spectrum value.

This is not a final answer. It is the shape a final answer would have to fill.

Definition 10C.3 — Blocker-profile projection

The **blocker-profile projection** is:

$$\pi_{\text{Bl}} : \mathfrak{Q}_T \rightarrow \mathfrak{Bl}_T.$$

It forgets non-generative diagnostic data and keeps the search-relevant obstruction structure:

$$\pi_{\text{Bl}}(\mathfrak{Q}_T) = (\Delta_T, \mathcal{A}_T, \tau_T, \chi_T, \mathcal{K}_T, \text{Need}_T).$$

Thus:

$$\mathfrak{Bl}_T = \pi_{\text{Bl}}(\mathfrak{Q}_T).$$

Definition 10C.4 — Question-answer mediation operator

The **question-answer mediation operator** is:

$$\text{M}_{\text{QA}} = \text{G}_{\text{gen}} \circ \pi_{\text{Bl}}.$$

That is:

$$\text{M}_{\text{QA}}(\mathfrak{Q}_T, \mathfrak{A}_U, \mathcal{K}_T) = \text{G}_{\text{gen}}(\pi_{\text{Bl}}(\mathfrak{Q}_T), \mathfrak{A}_U, \mathcal{K}_T).$$

It maps a diagnosed question-shape to a candidate-answer-shape.

Theorem 10C.5 — Question-to-Candidate-Answer Mediation Theorem

If BVD produces an admitted or conditional question-shape $\mathfrak{Q}_T$, and the blocker-profile projection $\pi_{\text{Bl}}(\mathfrak{Q}_T)$ admits at least one typed structural match to an unblocker packet in $\mathfrak{A}_U$, then Generative Bridge Search produces a candidate-answer-shape:

$$\mathfrak{A}_T^* = \text{M}_{\text{QA}}(\mathfrak{Q}_T, \mathfrak{A}_U, \mathcal{K}_T).$$

Proof

BVD supplies $\mathfrak{Q}_T$, including the diagnostic residual, active theorem/no-go subtree, branch field, status, and action need. The projection π_{Bl} extracts the blocker profile required for generative search. If an unblocker packet structurally matches that blocker profile, the generative search operator produces adapted candidate unblockers and verification obligations. These objects collectively form $\mathfrak{A}_T^*$. Therefore the mediation from question-shape to candidate-answer-shape is $\text{G}_{\text{gen}} \circ \pi_{\text{Bl}}$. $\square$

Corollary 10C.6 — Candidate-answer-shape is not admitted answer

$$\mathfrak{A}_T^* \not\Rightarrow \text{admitted solution.}$$

Admission requires:

$$\text{Adm}_{\mathcal{K}_T}(\mathfrak{A}_{T,k}^*) = 1$$

for at least one candidate, or an admitted atlas of partial candidates.

Corollary 10C.7 — BVD and GBS form a question-answer pair

At the methodology level:

$$\boxed{\text{BVD} : T \mapsto \mathfrak{Q}_T}$$

and:

$$\boxed{\text{GBS} : \mathfrak{Q}_T \mapsto \mathfrak{A}_T^*}$$

Thus BVD and GBS form a formal question/candidate-answer pair, mediated by the blocker profile.

Register note

It would be overclaimed to say:

$$\boxed{\text{GBS gives the answer.}}$$

The correct statement is:

$$\boxed{\text{GBS gives the candidate-answer-shape generated from the BVD question-shape.}}$$

10D. Reciprocal faces: BVD as question-shape, GBS as candidate-answer-shape

The generative score sheet must not be an independent rubric.

It must be derivable as a reciprocal face of BVD.

The shared primitive object is a bridge-problem state:

$$\boxed{\mathfrak{S}_T = (X_T, D_T, \mathcal{B}_T, \mathcal{A}_T, \Delta_T, \mathcal{K}_T, \mathcal{R}_T)}.$$

Where:

- X_T is the source presentation;
- D_T is the target distinction family;
- $\mathcal{B}_T$ is the branch field;
- $\mathcal{A}_T$ is the active theorem/no-go subtree;
- Δ_T is the residual family;
- $\mathcal{K}_T$ is the calibrator;
- $\mathcal{R}_T$ is the relation/trace data available to the programme.

BVD and GBS are two readout faces of this state.

Definition 10D.1 — Question face

The **question face** is the BVD readout:

$$\boxed{Q_{\text{BVD}}(\mathfrak{S}_T) = \mathfrak{Q}_T}$$

This extracts:

$$(\mathcal{D}_{\text{pre}}, \mathcal{D}_{\text{post}}, \Delta_{\text{diag}}, \text{Status}_T, \text{Act}_T).$$

Definition 10D.2 — Candidate-answer face

The **candidate-answer face** is the GBS readout:

$$A_{\text{GBS}}(\mathfrak{S}_T, \mathfrak{A}_U) = \mathfrak{A}_T^*.$$

This extracts or generates:

$$(\text{Need}_T, \{\mathfrak{U}_{T,k}^*\}, \{\text{Obl}_{T,k}^*\}, \{\Delta_{T,k}^*\}, \mathcal{K}_T^*, \text{Status}_{\text{gen}}).$$

Definition 10D.3 — Reciprocal-face mediation

BVD and GBS are reciprocal when:

$$A_{\text{GBS}} = G_{\text{gen}} \circ \pi_{\text{Bl}} \circ Q_{\text{BVD}}.$$

Thus the candidate-answer face is not independently selected.

It is generated by applying blocker projection and generative search to the BVD question face.

Theorem 10D.4 — Reciprocal-Face Theorem

BVD and Generative Bridge Search are reciprocal faces of the same RFOP bridge-problem state.

BVD reads:

what kind of question is this?

GBS reads:

what candidate-answer shapes does this question admit?

Proof

Both BVD and GBS begin from the same bridge-problem state $\mathfrak{S}_T$. BVD reads diagnostic demand, residuals, active subtrees, and action needs from that state, yielding $\mathfrak{Q}_T$. The blocker projection π_{Bl} extracts the generative obstruction from $\mathfrak{Q}_T$. GBS searches the source-admitted unblocker atlas and adapts matched unblockers to produce $\mathfrak{A}_T^*$. Therefore the GBS face is obtained from the BVD face by a derived mediation, not by an independent rubric. $\square$

Corollary 10D.5 — No independent generative rubric

The generative rubric must be a readout of:

1. BVD question-shape;
2. blocker-profile projection;
3. typed-match morphism;
4. adaptation core/shell split;
5. RBC verification obligations;
6. NUR/NUB/NUO/NFT guardrails.

If a generative score cannot be traced to these, it is not RFOP-admissible.

10E. Derivation of the generative score coordinates

This section derives the six generative score coordinates from the reciprocal-face structure.

10E.1 Coordinate derivation table

Generative coordinate	Derived from	Why it is forced
θ_{match}	BVD active subtree + blocker graph + triadic substitution	a candidate requires a typed match between source and target blocker forms
c_{core}	transfer core of Θ + source unblocker packet	a candidate must preserve some source mechanism core
s_{shell}^-	NFT + shell complement of Θ	non-preserved source assumptions must be redeclared or replaced
κ_{cand}	BVD repair target + candidate compatibility coverage	a candidate must say which target faces it covers
v_{obl}	RBC verification matrix	a candidate must state proof/construction/calibration obligations
g_{guard}	NUR/NUB/NUO/NFT guardrails	an unguarded candidate is inadmissible

No coordinate is optional. Removing one creates an untracked failure mode.

Theorem 10E.2 — Generative Score Pullback Theorem

The generative score vector:

$$\mathcal{G}_{\text{gen}} = (\theta_{\text{match}}, c_{\text{core}}, s_{\text{shell}}^-, \kappa_{\text{cand}}, v_{\text{obl}}, g_{\text{guard}})$$

is the pullback readout of the BVD question-shape against a source unblocker packet along a typed-match morphism.

Proof

The BVD question-shape supplies the target residual, active subtree, target-type jump, repair need, and action space. The source unblocker packet supplies a source blocker form, core mechanism, source shell, scope, and proof structure. The typed-match morphism Θ compares these structures. The comparison necessarily reads: how much blocker structure matches (θ_{match}); how much source mechanism core is preserved (c_{core}); what shell is not preserved (s_{shell}^-); what target coverage the adapted candidate supplies (κ_{cand}); what verification obligations are available (v_{obl}); and whether the guardrails pass (g_{guard}). These are exactly the coordinates of $\mathcal{G}_{\text{gen}}$. Therefore the score vector is derived as a pullback/comparison readout, not declared as an independent rubric. $\square$

Corollary 10E.3 — Status bands are quotients of the score readout

The status chart:

$$R < W < G < P < A$$

is a quotient of the finer viability readout:

$$\mathfrak{V}_{\text{gen}}$$

under the chosen granularity Γ_{gen} .

Thus the **existence** of a status readout is derived.

Only the **coarseness** of the chart is selected for audit convenience.

Corollary 10E.4 — Scoring disagreement is structural

A disagreement over a score is not a matter of taste. It must identify:

1. a different blocker graph;
2. a different typed-match morphism;
3. a different transfer-core/shell split;
4. a different compatibility coverage claim;
5. a different verification matrix;
6. a failed guardrail.

10F. Derivation of reconstruction-card fields

The reconstruction cards in §12B expose ten fields. This section derives each field.

Field	Derived from	Role
R_T	diagnostic seed table	identifies target problem row
Ω_T	BVD question face	gives formal question-shape
$\mathfrak{B}l_T$	$\pi_{\text{Bl}}(\Omega_T)$	gives blocker profile
R_j	source-admitted unblocker atlas / seed table	identifies source candidate packet
Θ	typed blocker-graph morphism	gives structural match
Adapt_Θ	RBC + triadic substitution + NFT	gives candidate-answer-shape
$\mathcal{G}_{\text{gen}}$	pullback score readout	gives candidate viability vector
V_T	RBC verification matrix	gives proof/calibration obligations
Guard	NUR/NUB/NUO/NFT	gives admissibility gates
Status	status quotient of $\mathcal{G}_{\text{gen}}$ plus admission gate	gives candidate maturity

Theorem 10F.1 — Reconstruction-Field Derivation Theorem

Every field in the worked-case reconstruction cards is derived from RFOP machinery.

Proof

The target row and question-shape are BVD outputs. The blocker profile is the blocker projection of the question-shape. The source row is drawn from the source-admitted unblocker atlas or seed table. The typed match is a blocker-graph morphism derived from triadic substitution and charted readout. Adaptation is the RBC-mediated update of the matched source core under target shell conditions. The score vector is the pullback readout of the BVD question-shape against the source unblocker packet. The verification matrix is required by RBC and no-undeclared boundary/observer discipline. The guard field is supplied by NUR/NUB/NUO/NFT. The status is the quotient of the score readout through Γ_{gen} , with admitted status gated by $\text{Adm}_{\mathcal{K}_T}$. Therefore every field is programme-derived. $\square$

Corollary 10F.2 — Reconstruction cards are not bookkeeping

The reconstruction cards are not presentation aids. They are the minimal visible trace of the programme-derived generation pipeline.

10G. Source-admitted unblocker atlas population

Earlier versions defined the schema:

$$\mathfrak{A}_U^{\text{adm}}$$

but did not yet populate it concretely.

10G.1 Canonical unblocker families

Unblocker family	Structural form solved	Candidate transfer core
Stone duality	algebraic/topological presentation equivalence	spectrum / ultrafilter reconstruction
Gelfand duality	commutative algebra $\leftrightarrow$ compact space	operator-spectrum geometry
Spectral reconstruction	operator spectrum encodes structure	invariant/operator readout
Spec/sheaf reconstruction	local algebra patches to geometry	local-to-global gluing
Noether-style bridges	symmetry $\leftrightarrow$ invariant structure	compatibility-law generation
RG/EFT matching	multi-scale compatibility	scale-local matching
Empirical-calibrator packet	formal model $\leftrightarrow$ observable/readout	likelihood/admission bridge
Causal-inference packet	intervention/readout separation	explicit bridge assumptions

10G.1A Source-status boundary for atlas families

The families in §10G.1 are canonical **unblocker-family types**.

They are not automatically admitted theorem instances.

A family becomes a source-admitted packet only when an application paper supplies:

specific theorem/source + hypotheses + solved blocker profile + transfer core + non-transfer shell.

Thus:

Stone duality as a family $\neq$ a source-admitted Stone packet for a target problem.

This prevents the atlas from smuggling external theorems without source cards.

10G.1B First instance-level unblocker atlas table

The following table is not an exhaustive atlas. It is the first populated instance layer. Each row still requires source-card expansion in an application paper before being used as a load-bearing theorem packet.

Family	Explicit instance	Solved blocker profile	Transfer core	Non-transfer shell
topological duality	Stone duality for Boolean algebras and Stone spaces	algebraic presentation $\leftrightarrow$ topological presentation	spectrum/ultrafilter reconstruction	Boolean/distributive assumptions do not transfer automatically
operator/topology duality	Gelfand duality for commutative C*-algebras	algebraic observables $\leftrightarrow$ compact Hausdorff space	spectrum-as-space construction	commutativity, norm, *-algebra hypotheses
spectral reconstruction	spectral theorem / inverse spectral patterns	operator data constrains structure	invariant/operator spectrum	spectral completeness rarely automatic
local-to-global geometry	sheaf / gluing / Spec-style reconstruction	local algebraic data to global space	local compatibility + gluing	sheaf, locality, and cover hypotheses
symmetry bridge	Noether-style theorem	symmetry/action relation to invariant/conservation	symmetry-to-invariant mechanism	action/differentiability/variational hypotheses
scale bridge	RG/EFT matching	multiple scale descriptions compatible by flow/matching	scale-local compatibility	regulator, regime, and matching hypotheses
statistical bridge	Bayesian likelihood / statistical calibration	model to data-admission bridge	likelihood/update/admission	model/data assumptions
causal bridge	structural causal model / do-calculus pattern	association-to-intervention transfer blocked without assumptions	explicit intervention bridge	causal sufficiency/graph assumptions
empirical bridge	measurement model + instrument likelihood	formal target to observable readout	model-data calibration	instrument/noise/data regime
computational bridge	algorithmic reduction / representation theorem	hard target transformed to tractable presentation	reduction/encoding equivalence	complexity/model-of-computation assumptions

This table closes the manuscript-level need for a populated atlas seed. It does not close the application-level need for source cards.

Definition 10G.2 — Atlas population rule

A source family enters the source-admitted unblocker atlas only if:

1. the source theorem/construction is externally admitted or source-grounded;
2. the solved blocker form is identifiable;
3. the transferable core can be separated from the source shell;
4. the non-transfer boundary is explicit.

10H. Derived blocker-shape similarity

The paper must not leave unblocker selection to intuition.

Definition 10H.1 — Structural similarity profile

For target blocker:

$$\mathfrak{Bl}_T$$

and source blocker:

$$\mathfrak{Bl}_j,$$

define:

$$\Sigma(\mathfrak{Bl}_j, \mathfrak{Bl}_T) = (\sigma_{\text{face}}, \sigma_{\text{transfer}}, \sigma_{\text{need}}, \sigma_{\text{cal}}, \sigma_{\text{dep}}).$$

Where:

Coordinate	Meaning
σ_{face}	overlap of active face structure
σ_{transfer}	overlap of target-type jump structure
σ_{need}	overlap of unblocker need
σ_{cal}	overlap of calibrator/readout role
σ_{dep}	overlap of dependency structure

Definition 10H.2 — Structural similarity operator

$$\text{Sim}_{\text{Bl}} : (\mathfrak{Bl}_j, \mathfrak{Bl}_T) \mapsto \Sigma(\mathfrak{Bl}_j, \mathfrak{Bl}_T).$$

This is not semantic similarity.

It is typed obstruction-structure comparison.

Theorem 10H.3 — Unblocker Selection Derivation Theorem

Unblocker selection is derived from blocker-shape comparison and adaptation structure, not from analogy or expert preference.

Proof

The target blocker graph determines the required face structure, target-type jumps, unblocker need, calibrator role, and dependency structure. A source unblocker packet is eligible only if its solved blocker graph admits a typed-match morphism into the target blocker graph. The similarity operator records this preservation. Candidate ranking then depends on transfer-core preservation, shell burden, compatibility coverage, verification obligations, and guardrail satisfaction. Therefore unblocker selection is determined by derived blocker-shape comparison and adaptation structure rather than intuitive resemblance. $\square$

Corollary 10H.4 — Reader reproducibility of unblocker search

A reader may dispute unblocker selection only by disputing:

1. blocker graph;
2. typed-match morphism;
3. similarity profile;
4. core/shell split;
5. compatibility coverage;
6. verification obligations;
7. guardrail status.

11. Diagnostic tables as generative seed sets

Bridge-Valence Diagnostics and the CST diagnostic atlas are analytical papers. Their rows classify problems.

Generative Bridge Search uses those rows differently.

A diagnostic row can function as:

1. a **target blocker profile** to be solved or reduced;
2. a **source blocker profile** whose structure may match another problem;
3. an **unblocker-source row** if the row contains a known compatibility operator;
4. a **negative guardrail row** showing what not to transfer.

Thus the broad and deep diagnostic tables become search indices for candidate generation.

Definition 11.1 — Generative diagnostic seed row

A diagnostic row R_i is a **generative seed row** when it supplies:

$$R_i = (D_i, \mathcal{D}_i, \mathcal{A}_i, C_i, \Delta_i, a_i)$$

where:

- D_i is the declared target regime;
- $\mathcal{D}_i = (\rho, b, s, n, \kappa, h)$ is the diagnostic trace;
- $\mathcal{A}_i$ is the active theorem/no-go subtree;
- C_i is the diagnostic class;
- Δ_i is the residual/guardrail;
- a_i is the action implication.

The generative search reads R_i as either:

$$\mathfrak{B}l_i$$

a blocker profile, or:

$$\mathfrak{U}_i$$

an unblocker packet source, depending on whether C_i is critical/residualized or finite/compatibility-stabilized.

11.2 Broad diagnostic seed table from BVD

This table compresses the 36-case BVD landscape into a generative index.

#	BVD case	Diagnostic class	Generative role
1	Classical harmonic oscillator	bounded bridge	negative control: no generator needed beyond local closure
2	Kepler / Newtonian two-body	bounded bridge	source of bounded-model discipline and perturbation boundary
3	Newtonian mechanics	bounded bridge	source of regime-boundary guardrail
4	Heat equation / diffusion	bounded / conditional	source of boundary-condition / PDE-calibrator pattern
5	Maxwellian electromagnetism	finite unification atlas	unblocker-source: compatibility operator collapses high-valence field phenomena
6	Special relativity	finite unification atlas	unblocker-source: invariant compatibility structure
7	General relativity	finite unification / conditional	unblocker-source inside classical metric-gravity target; guardrail outside target
8	Electroweak unification	finite unification / conditional	unblocker-source: symmetry breaking / gauge compatibility
9	Standard Model gauge structure	high-valence finite unification	unblocker-source: gauge/representation compatibility pattern
10	RG / EFT matching	finite bridge atlas / conditional	unblocker-source: scale-local compatibility and matching
11	Thermodynamics to statistical mechanics	mixed / conditional finite atlas	blocker/unblocker mixed: limit, typicality, micro-macro bridge
12	Schrödinger dynamics	bounded inside target	guardrail: bounded target excludes measurement
13	Quantum measurement	critical or mixed	target blocker: observer/readout/outcome/update compatibility
14	Black-hole thermodynamics / holography	critical / finite-atlas borderline	target blocker: dictionary/reconstruction/dynamics
15	Causal-set faithful embedding	critical or conditional atlas	target blocker / CST bridge source

#	BVD case	Diagnostic class	Generative role
16	Observer-slice AC node	critical relation locus	target blocker for RFOP continuum branch
17	Quantum gravity / spacetime emergence	critical / underdetermined	target blocker family
18	Empirical physical spacetime recovery	critical / conditional	empirical-readout blocker
19	Rigid-body mechanics	bounded-to-finite atlas	source: constrained-system compatibility
20	Classical optics	bounded / conditional	source: approximation-regime bridge
21	Lagrangian/Hamiltonian equivalence	bounded / conditional	source: transform under regularity
22	Noether theorem	finite bridge theorem	unblocker-source: symmetry-action-conservation bridge
23	Navier-Stokes local	bounded / conditional	source: local-function-space boundary
24	Navier-Stokes global	critical / underdetermined	target blocker: global a priori estimate
25	Turbulence closure	critical / mixed	target blocker: scale cascade / closure model
26	Ising phase transition	finite unification / conditional	source: partition/RG compatibility
27	Critical phenomena	finite-to-critical atlas / mixed	source/target: universality fixed-point atlas
28	Bayesian inference	bounded bridge	source: prior/likelihood/model calibrator
29	Causal inference	conditional bridge atlas	source/target: no-free-transfer association-to-intervention
30	ML prediction	bounded / conditional	source: fixed-distribution evaluator/loss
31	OOD generalization / agency	critical / mixed	target blocker: invariance under shift
32	Evolution by natural selection	finite unification / conditional	source: variation/inheritance/selection dynamics
33	Abiogenesis	critical / underdetermined	target blocker: chemistry-to-life transition
34	Consciousness	critical / underdetermined	target blocker: first/third-person readout
35	Quantum measurement detailed	critical / mixed	target blocker: interpretation-specific bridges

#	BVD case	Diagnostic class	Generative role
36	Formal theory to empirical science	critical / conditional-mixed	target blocker and empirical-calibrator source

11.3 Deep CST diagnostic seed table

This table compresses the CST diagnostic atlas into a generative index.

#	CST component	Diagnostic class	Generative role
1	partial order primitive	bounded bridge	source primitive: order/proto-causality
2	local finiteness	bounded bridge	source primitive: finite interval/discreteness
3	order + number slogan	finite/conditional atlas	blocker guardrail: order+count not full geometry
4	Poisson sprinkling	conditional bridge	source: continuum-to-causet sampling shell
5	BHS Lorentz-invariance protection	finite compatibility bridge	unblocker-source: no-preferred-frame compatibility
6	faithful embedding	conditional / residualized	target/source: embedding and closeness profile
7	Hauptvermutung	critical / theorem-sensitive	target blocker: uniqueness/similarity operator
8	manifoldlikeness recognition	critical / underdetermined	target blocker: classifier/calibrator need
9	Myrheim-Meyer dimension	conditional bounded bridge	source: dimension-estimator pattern
10	midpoint-scaling dimension	conditional bounded bridge	source: chain/count scaling pattern
11	topology via thickened antichains	conditional finite atlas	source: topology/homology recovery pattern
12	stable homology indicator	conditional / residualized	source: stability evidence, not sufficiency
13	proper-time / geodesic length	conditional bounded bridge	source: chain-length/proper-time bridge
14	causal-set d'Alembertian	finite-to-critical atlas	source: operator-recovery candidate
15	Benincasa-Dowker scalar curvature/action	conditional finite atlas	source: operator/action recovery pattern
16	classical sequential growth	finite/conditional dynamics atlas	source: covariance/growth dynamics shell

#	CST component	Diagnostic class	Generative role
17	quantum dynamics / sum over causet	critical / underdetermined	target blocker: measure/continuum dominance
18	empirical phenomenology	critical / conditional	target blocker: model-to-observable calibrator

11.4 How seed rows become worked examples

A worked example starts by selecting:

1. a target row R_T with critical/residualized status;
2. one or more source rows R_j with finite/conditional compatibility mechanisms;
3. a proposed typed match:

$$\Theta : R_j \rightarrow R_T;$$

4. a candidate adapted unblocker:

$$\text{Adapt}_\Theta(\mathcal{U}_j, T);$$

5. a verification matrix.

The tables therefore become the search base for the generative procedure.

12. End-to-end worked generative traces

The examples in this section must carry the full methodology.

A worked example is complete only if it includes:

$$R_T \rightarrow \mathfrak{B}l_T \rightarrow \{R_j\} \rightarrow \Theta \rightarrow \text{Adapt}_\Theta \rightarrow \text{Obl}_T^* \rightarrow \text{Guard} \rightarrow \text{Status}.$$

Where:

- R_T is the target diagnostic seed row;
- $\mathfrak{B}l_T$ is the target blocker profile;
- $\{R_j\}$ are source candidate rows;
- Θ is the typed structural match;
- Adapt_Θ is the candidate adaptation;
- Obl_T^* are verification obligations;
- Guard records guardrail checks;
- Status says generated, rejected, pending, or admitted.

12.1 Negative control: bounded rows do not trigger generation

Target seed row

$$R_T = \text{BVD row 1: classical harmonic oscillator.}$$

Class:

bounded bridge.

Blocker profile

There is no critical blocker profile. The declared target already has a compatibility law:

$$m\ddot{x} + kx = 0.$$

Generative decision

CompatSearch is not activated.

Guardrail result

This row tests against over-generation. The method must not inflate bounded problems into critical atlases.

Status

no generative search required.

12.2 Maxwell row as an unblocker-source pattern

Source seed row

R_j = BVD row 5: Maxwellian electromagnetism.

Class:

finite unification atlas.

Source blocker profile

$\mathfrak{B}_{\text{preEM}} = (\Delta_{\text{electric/magnetic}}, \mathcal{A}_{\text{field}}, \tau_{\text{field/source/wave}}, \chi_{\text{coupled fields}}, \mathcal{K}_{\text{classical}}, \text{Need}_{\text{compatibility}}).$

Unblocker packet

$\mathfrak{U}_{\text{Maxwell}} = (P_{\text{EM}}, \mathfrak{B}_{\text{preEM}}, \kappa_{\text{Maxwell}}, \mathcal{K}_{\text{EM}}, \Pi_{\text{field}}, \text{Scope}_{\text{classical}}).$

Possible target class

A target problem may use this unblocker only if its blocker form is:

many coupled field-like target faces lacking one compatibility law.

Guardrail check

Guardrail	Result
typed blocker match	required; not automatic
no import without adaptation	Maxwell equations do not transfer directly
verification obligations	must state target-side compatibility equations
novelty trace	source row and adaptation must be recorded
NFT check	field compatibility does not imply dynamics/empirical closure

Status

source unblocker pattern, not direct transferable theorem.

12.3 CST Hauptvermutung: end-to-end candidate generation

Target seed row

$$R_T = \text{CST row 7: Hauptvermutung.}$$

Class:

critical / theorem-sensitive.

Target blocker profile

$$\mathfrak{B}_{\text{Haupt}} = (\Delta_{\text{uniqueness}}, \Delta_{\text{similarity}}, \Delta_{\text{topology}}, \Delta_{\text{metric}}, \Delta_{\text{statistical}}, \mathcal{K}_{\text{close}}, \text{Need}_{\text{reconstruction}}).$$

Source candidate rows

Source row	Candidate mechanism
CST row 11: topology via thickened antichains	conditional topology/homology recovery
CST row 14: causal-set d'Alembertian	operator recovery from order/count
CST row 15: Benincasa-Dowker action	action/curvature readout
BVD row 22: Noether theorem	invariant/conservation from symmetry-action bridge
BVD row 29: causal inference	explicit assumptions for no-free-transfer promotion

Typed match

The proposed match is not vocabulary-level. It is:

$$\Theta_{\text{rec}} : \text{conditional reconstruction mechanism} \rightarrow \text{continuum-similarity operator need.}$$

Preserved structure:

1. source data do not directly determine target;
2. extra compatibility conditions are required;
3. a constructed invariant/operator may control a target comparison;
4. proof obligation is a sufficiency theorem, not mere correlation.

Adaptation output

The generated candidate family is:

$$\mathcal{G}_{\text{gen}}(\mathfrak{B}_{\text{Haupt}}) = \{\mathfrak{U}_{\text{top}}^*, \mathfrak{U}_{\square}^*, \mathfrak{U}_{\text{action}}^*, \mathfrak{U}_{\text{inv}}^*\}.$$

Where, for example:

$$\mathfrak{U}_{\square}^* = (\kappa_{\square}^{\text{Haupt}}, \mathcal{K}_{\text{close}}^{\square}, \Pi_{\text{spectral/similarity}}, \Delta_{\text{residual}}^{\square}, \text{Obl}_{\square}^*).$$

Verification obligations

Candidate	Required verification
$\mathcal{U}_{\text{top}}^*$	prove topology/homology recovery controls enough of continuum similarity
$\mathcal{U}_{\square}^*$	prove operator spectrum/invariants determine target class to tolerance
$\mathcal{U}_{\text{action}}^*$	prove action/curvature readout distinguishes non-similar continuum targets
$\mathcal{U}_{\text{inv}}^*$	prove invariant completeness or state residual incompleteness

Guardrail result

Guardrail	Result
typed blocker match	partial; requires declared similarity target
no import without adaptation	satisfied only if source shell assumptions are redeclared
verification obligations	explicit
novelty trace	source rows recorded
NFT check	active: topology/operator/action do not automatically imply continuum uniqueness

Status

candidate family generated; no candidate admitted yet.

12.4 RFOP GR/AC/continuum: end-to-end candidate generation

Target seed rows

$R_T = \text{BVD row 16: observer-slice AC node}$

and:

$R'_T = \text{RFOP self-audit row 13: GR/AC/continuum.}$

Class:

critical relation locus.

Target blocker profile

$\mathfrak{B}_{\text{GR}} = (\Delta_{\text{AC}}, \Delta_{\text{cont}}, \Delta_{\text{manifold}}, \Delta_{\text{dim}}, \Delta_{\text{conf}}, \Delta_{\text{metric}}, \Delta_{\text{dyn}}, \Delta_{\text{emp}}, \mathcal{K}_{\text{GR}}, \text{Need}_{\text{compatibility}}).$

Source candidate rows

Source row	Candidate mechanism
CST row 11	topology/homology recovery through thickened regions
CST row 14	operator recovery from causal/order structure
CST row 15	action/curvature readout
BVD row 6	invariant interval compatibility
BVD row 22	symmetry-action-conservation bridge
BVD row 36	formal-to-empirical calibrator

Typed matches

Match	Preserved structure	Risk
Θ_{top}	region system to topological readout	homology not manifold recovery
$\Theta_{\square}$	causal/order data to operator-like readout	operator not dynamics
Θ_{action}	action-like readout from region/causal data	action-like not physical action
Θ_{inv}	invariant compatibility target	invariant not full metric/dynamics
Θ_{emp}	formal structure to empirical readout	empirical calibrator not supplied

Adaptation outputs

$$G_{\text{gen}}(\mathcal{B}_{\text{GR}}) = \{\mathcal{U}_{\text{top}}^*, \mathcal{U}_{\square}^*, \mathcal{U}_{\text{action}}^*, \mathcal{U}_{\text{inv}}^*, \mathcal{U}_{\text{emp}}^*\}.$$

Verification matrix

Candidate	Verification obligation	No-free-transfer warning
$\mathcal{U}_{\text{top}}^*$	define RFOP region thickening/nerve/homology and prove stability	topology does not imply metric
$\mathcal{U}_{\square}^*$	define operator-like readout and prove continuum relevance	operator does not imply dynamics
$\mathcal{U}_{\text{action}}^*$	define action-like functional and target scope	action-like not physical action
$\mathcal{U}_{\text{inv}}^*$	prove invariant controls target face	invariant may be incomplete
$\mathcal{U}_{\text{emp}}^*$	declare measurement/model/statistical calibrator	empirical readout not formal closure

Guardrail result

All candidates remain:

generated / pending verification.

None is admitted as a GR derivation.

Status

candidate compatibility-search portfolio generated.

This is genuine generation, but not solution.

12.5 Quantum measurement blocker family

Target seed rows

$$R_T = \text{BVD rows 13 and 35.}$$

Class:

critical / mixed.

Target blocker profile

$$\mathfrak{B}_{\text{meas}} = (\Delta_{\text{outcome}}, \Delta_{\text{update}}, \Delta_{\text{probability}}, \Delta_{\text{observer}}, \Delta_{\text{empirical}}, \mathcal{K}_{\text{meas}}, \text{Need}_{\text{outcome/readout}}).$$

Source candidate rows

Source row	Candidate mechanism
BVD row 12: Schrödinger dynamics	bounded dynamics, shows what is insufficient
BVD row 36: formal-to-empirical science	empirical calibrator template
BVD row 29: causal inference	intervention/readout separation
BVD row 11: thermodynamics/stat-mech	micro-macro / typicality bridge
BVD row 31: OOD/generalization	context-shift/evaluator stability

Generated output

candidate atlas of measurement subbridges

rather than one measurement solution.

Subbridges:

dynamics, outcome, probability, observer/readout, empirical calibration.

Guardrail result

A single imported mechanism fails unless it declares how it handles all target faces. The generated result is a candidate atlas, not a terminal interpretation.

12.6 Empirical bridge generation from BVD row 36

Target/source row

R = BVD row 36: formal theory to empirical science.

This row can function as both:

1. a target blocker for formal theories lacking empirical readout;
2. a source unblocker template for empirical calibration.

Unblocker packet template

$$\mathfrak{U}_{\text{emp}} = (\text{formal-to-empirical bridge}, \mathfrak{B}_{\text{emp}}, \kappa_{\text{model/data}}, \mathcal{K}_{\text{emp}}, \Pi_{\text{validation}}, \text{Scope}_{\text{empirical}}).$$

Adaptation output

For any formal branch T , the adapted candidate is:

$$\text{Adapt}_{\text{emp}}(\mathfrak{U}_{\text{emp}}, T) = (\text{model map}, \text{observable readout}, \text{data likelihood}, \text{admission criterion}, \text{update rule}).$$

Verification obligations

1. define observable;
2. define instrument/readout;
3. define model-to-data map;
4. define uncertainty/statistical test;
5. define admission/update rule.

Status

empirical-calibrator candidate generated.

Not empirical prediction until data/model verification is supplied.

12A. Generative decision matrices for the worked examples

This section applies the generative score sheet to the worked examples.

The scores are ordinal and target-relative:

$$L < M < H < VH.$$

They are not truth values. They determine whether a candidate is rejected, weak, generated pending, high-priority, or admitted.

12A.1 Hauptvermutung candidate matrix

Target:

$$R_T = \text{CST row 7: Hauptvermutung.}$$

Candidate	θ_{match}	c_{core}	s_{shell}^-	κ_{cand}	v_{obl}	g_{guard}	Status
topology/homology recovery candidate	H	H	H	M/H	H	VH	generated pending, shell- heavy
operator/spectral candidate	H	H	H	H	H	VH	generated pending
action/curvature candidate	M/H	H	VH	M/H	H	VH	weak/generated pending, high shell burden
invariant- comparison candidate	H	M/H	H	H	H	VH	generated pending
causal- inference assump- tion bridge	M	M	M	M	H	VH	weak method- ological candidate, not direct solution

Interpretation:

operator/spectral and invariant-comparison candidates are best first verification targets.
--

But none is admitted.

12A.2 RFOP GR / AC / continuum candidate matrix

Target:

$$R_T = \text{RFOP GR/AC/continuum.}$$

Candidate	θ_{match}	c_{core}	s_{shell}^-	κ_{cand}	v_{obl}	g_{guard}	Status
topology/homology	H	H	H	M/H	H	VH	generated
readout	H	H	H	H	H	VH	pending
operator-like							generated
readout							pending, high priority
action-like	M/H	H	VH	M	H	VH	weak/generated
functional							pending; overclaim risk
invariant interval target	M	M	H	M	M	VH	weak candidate
empirical calibrator	H	H	H	M/H	VH	VH	generated pending for empirical branch only

Interpretation:

operator-like readout and empirical calibrator are highest-priority candidate tracks, but they verify different targets.
--

The operator-like candidate does not verify dynamics. The empirical calibrator does not verify formal continuum recovery.

12A.3 Quantum measurement candidate matrix

Target:

$$R_T = \text{BVD rows 13 and 35.}$$

Candidate	θ_{match}	c_{core}	s_{shell}^-	κ_{cand}	v_{obl}	g_{guard}	Status
Schrödinger dynamics extension	M	H	H	M	M	VH	negative/weak: shows insufficiency

Candidate	θ_{match}	c_{core}	s_{shell}^-	κ_{cand}	v_{obl}	g_{guard}	Status
formal-to-empirical calibrator	H	H	H	M/H	VH	VH	generated pending for readout face
causal-inference intervention bridge	H	M/H	H	M/H	H	VH	generated pending for intervention/update face
micro-macro typicality bridge	H	H	VH	M/H	H	VH	weak/generated pending; shell-heavy
OOD/evaluation stability bridge	M/H	M	H	M	H	VH	weak methodological candidate

Interpretation:

the method generates a measurement atlas, not a single measurement solution.

12A.4 Empirical bridge candidate matrix

Target/source:

R = BVD row 36: formal theory to empirical science.

Candidate template	θ_{match}	c_{core}	s_{shell}^-	κ_{cand}	v_{obl}	g_{guard}	Status
model-to-observable map	H	H	H	H	VH	VH	generated pending
instrument/readout map	H	H	H	H	VH	VH	generated pending
likelihood/statistical calibrator	H	H	M/H	H	VH	VH	high-priority empirical candidate
intervention/validation protocol	M/H	M/H	H	M/H	H	VH	generated pending

Interpretation:

empirical bridge generation is reproducible because the calibrator components are explicit.

Theorem 12A.5 — Worked-Example Decision Reproducibility Theorem

The worked examples in §12 are reproducible at BVD-derived generative granularity Γ_{gen} because every candidate is assigned a decision vector:

$$(\theta_{\text{match}}, c_{\text{core}}, s_{\text{shell}}^-, \kappa_{\text{cand}}, v_{\text{obl}}, g_{\text{guard}})$$

and a status determined by the candidate status rule.

Proof

Each example declares a target seed row, source candidate row, blocker profile, candidate mechanism, adaptation burden, verification obligations, and guardrail result. These are precisely the inputs of $\mathcal{G}_{\text{gen}}$. The candidate status rule maps $\mathcal{G}_{\text{gen}}$ to rejected, weak, generated pending, high-priority, or admitted status. Therefore a reader can reproduce or challenge each example by changing a coordinate or row declaration. $\square$

12B. Worked-case reconstruction cards

This section checks whether the worked examples are exactly aligned to the paper’s machinery.

A worked case is complete only if it exposes the same ten fields:

$$R_T, \quad \Omega_T, \quad \mathfrak{Bl}_T, \quad R_j, \quad \Theta, \quad \text{Adapt}_\Theta, \quad \mathcal{G}_{\text{gen}}, \quad V_T, \quad \text{Guard}, \quad \text{Status}.$$

These are not optional explanatory aids. They are the reconstruction card that lets a reader reproduce the example.

12B.1 Reconstruction-card template

Field	Required content
R_T	target diagnostic seed row
Ω_T	BVD question-shape
$\mathfrak{Bl}_T$	blocker profile obtained by $\pi_{\text{Bl}}(\Omega_T)$
R_j	source candidate row(s)
Θ	typed blocker-graph match
Adapt_Θ	candidate-answer-shape / adapted candidate
$\mathcal{G}_{\text{gen}}$	generative decision vector
V_T	verification matrix
Guard	five guardrail results
Status	rejected / weak / pending / high-priority / admitted

12B.2 Card A — bounded-row negative control

Field	Value
R_T	BVD row 1: classical harmonic oscillator
Ω_T	bounded classical mass-spring target with strong equation law
$\mathfrak{Bl}_T$	no critical blocker; compatibility law already present
R_j	none required

Field	Value
Θ	none
Adapt_Θ	none
$\mathcal{G}_{\text{gen}}$	not activated
V_T	standard target verification of the oscillator equation
Guard	over-generation guard passes by refusing search
Status	no generative search required

Reader check:

bounded bridge $\Rightarrow$ do not force generative candidate search.

12B.3 Card B — Maxwell source-unblocker pattern

Field	Value
R_T	not a target case here; Maxwell is source row applies only after a target high-valence field-like blocker is selected
Ω_T	
$\mathfrak{B}l_T$	target must have coupled field-like target faces lacking common law
R_j	BVD row 5: Maxwellian electromagnetism
Θ	allowed only if target blocker graph preserves coupled-field compatibility need
Adapt_Θ	target-side compatibility-law candidate, not Maxwell equations imported
$\mathcal{G}_{\text{gen}}$	depends on selected target; not scored globally
V_T	must define target field variables, equations, residuals, and failure conditions
Guard	no import without adaptation; source theorem does not transfer
Status	source-unblocker packet schema, not generated candidate by itself

Reader check:

source-unblocker row $\neq$ target candidate until R_T is declared.

12B.4 Card C — CST Hauptvermutung

Field	Value
R_T	CST row 7: Hauptvermutung critical/theorem-sensitive uniqueness/similarity question
Ω_T	
$\mathfrak{B}l_T$	$(\Delta_{\text{uniqueness}}, \Delta_{\text{similarity}}, \Delta_{\text{topology}}, \Delta_{\text{metric}}, \Delta_{\text{statistical}}, \mathcal{K}_{\text{close}}, \text{Nee})$
R_j	CST rows 11, 14, 15; BVD rows 22, 29
Θ	Θ_{rec} : conditional reconstruction mechanism $\rightarrow$ continuum-similarity operator need
Adapt_Θ	$\{\mathcal{U}_{\text{top}}^*, \mathcal{U}_{\square}^*, \mathcal{U}_{\text{action}}^*, \mathcal{U}_{\text{inv}}^*\}$
$\mathcal{G}_{\text{gen}}$	decision matrix in §12A.1

Field	Value
V_T	prove invariant/operator/topology/action controls declared similarity target
Guard Status	all pass as candidates; NFT remains active generated pending; no candidate admitted

Reader check:

The reader can challenge this case by denying one of:

1. the target row classification;
2. the blocker profile;
3. the reconstruction match Θ_{rec} ;
4. the decision vector in §12A.1;
5. the verification obligations.

12B.5 Card D — RFOP GR / AC / continuum

Field	Value
R_T $\mathfrak{Q}_T$	BVD row 16 and RFOP self-audit row 13 critical relation locus: observer-slice AC / continuum / causal-region metric target
$\mathfrak{Bl}_T$ R_j Θ	$(\Delta_{\text{AC}}, \Delta_{\text{cont}}, \Delta_{\text{manifold}}, \Delta_{\text{dim}}, \Delta_{\text{conf}}, \Delta_{\text{metric}}, \Delta_{\text{dyn}}, \Delta_{\text{emp}}, \mathcal{K}_{\text{GR}}, \mathcal{N})$ CST rows 11, 14, 15; BVD rows 6, 22, 36
Adapt_Θ $\mathcal{G}_{\text{gen}}$ V_T	$\Theta_{\text{top}}, \Theta_{\square}, \Theta_{\text{action}}, \Theta_{\text{inv}}, \Theta_{\text{emp}}$ $\{\mathfrak{U}_{\text{top}}^*, \mathfrak{U}_{\square}^*, \mathfrak{U}_{\text{action}}^*, \mathfrak{U}_{\text{inv}}^*, \mathfrak{U}_{\text{emp}}^*\}$ decision matrix in §12A.2
Guard	topology/operator/action/invariant/empirical verification matrices
Status	all candidates pending; no GR/dynamics claim admitted candidate compatibility-search portfolio generated

Reader check:

operator-like candidate $\nRightarrow$ dynamics;

topology-like candidate $\nRightarrow$ metric.

The worked case is aligned if those non-transfer boundaries remain visible.

12B.6 Card E — quantum measurement

Field	Value
R_T $\mathfrak{Q}_T$	BVD rows 13 and 35 critical/mixed bridge from quantum formalism to definite outcome/readout/update
$\mathfrak{Bl}_T$ R_j	$(\Delta_{\text{outcome}}, \Delta_{\text{update}}, \Delta_{\text{probability}}, \Delta_{\text{observer}}, \Delta_{\text{empirical}}, \mathcal{K}_{\text{meas}}, \mathcal{N}_{\text{meas}})$ BVD rows 12, 36, 29, 11, 31

Field	Value
Θ	subface matches: dynamics, empirical readout, intervention/update, micro-macro, evaluator stability
Adapt_Θ	candidate measurement subbridge atlas
$\mathcal{G}_{\text{gen}}$	decision matrix in §12A.3
V_T	separate verification matrices for outcome, update, probability, observer, empirical faces
Guard	single-solution transfer blocked
Status	generated atlas, not measurement solution

Reader check:

The case is aligned because it refuses:

Schrödinger dynamics $\Rightarrow$ measurement solution.

12B.7 Card F — empirical bridge generation

Field	Value
R_T	BVD row 36
$\mathfrak{Q}_T$	formal-theory-to-empirical-science bridge shape
$\mathfrak{Bl}_T$	$(\Delta_{\text{model}}, \Delta_{\text{observable}}, \Delta_{\text{instrument}}, \Delta_{\text{statistics}}, \Delta_{\text{validation}}, \mathcal{K}_{\text{emp}}, \text{Nee})$
R_j	BVD row 36 as reusable source template
Θ	formal-to-empirical calibrator shape $\rightarrow$ target formal branch
Adapt_Θ	model map, observable readout, data likelihood, admission criterion, update rule
$\mathcal{G}_{\text{gen}}$	decision matrix in §12A.4
V_T	observable/instrument/statistical/admission matrix
Guard	no empirical prediction without data/model calibrator
Status	empirical-calibrator candidate generated

Reader check:

formal structure $\nRightarrow$ empirical prediction.

12B.8 Worked-case alignment theorem

Theorem 12B.8 — Worked-Case Machinery Alignment Theorem

The worked cases in §12 are aligned to the paper’s machinery if and only if each case admits a reconstruction card containing:

$$R_T, \mathfrak{Q}_T, \mathfrak{Bl}_T, R_j, \Theta, \text{Adapt}_\Theta, \mathcal{G}_{\text{gen}}, V_T, \text{Guard}, \text{Status}.$$

Proof

These ten fields are exactly the objects required by the paper’s pipeline: BVD supplies R_T and $\mathfrak{Q}_T$; blocker projection supplies $\mathfrak{Bl}_T$; seed tables supply R_j ; typed matching supplies Θ ; adaptation supplies

Adapt_Θ; the generative score sheet supplies $\mathcal{G}_{\text{gen}}$; RBC supplies V_T ; NUR/NUB/NUO/NFT supply Guard; and the status spectrum supplies Status. If any field is missing, the case no longer carries the full methodology. If all fields are present, the case is reconstructible. $\square$

12B.9 Remaining reader ambiguity

After reconstruction cards, the main reader ambiguities are localized to:

1. whether a source row should have been selected;
2. whether the typed match score is too generous;
3. whether shell burden is undercounted;
4. whether candidate compatibility coverage is overstated.

These are no longer global ambiguities. They are row-level challenge points.

12C. Score provenance cards

The decision matrices in §12A are not meant to be taken on authority. This section shows how a reader can reconstruct representative scores.

12C.1 Hauptvermutung operator/spectral candidate

Target:

$$R_T = \text{CST row 7.}$$

Source:

$$R_j = \text{CST row 14: causal-set d'Alembertian/operator recovery.}$$

Coordinate	Score	Derivation
θ_{match}	H	preserves source-data $\rightarrow$ operator/invariant $\rightarrow$ target-comparison need, but not full uniqueness
c_{core}	H	operator recovery core is preserved as candidate invariant mechanism
s_{shell}^-	H	source operator assumptions must be replaced by Hauptvermutung closeness assumptions
κ_{cand}	H	candidate could cover metric/similarity target partially, not topology/statistics fully
v_{obl}	H	proof obligations are clear: invariant stability, sufficiency, target completeness
g_{guard}	VH	all guardrails pass as candidate, not solution

Status:

generated pending.

Reason:

$$\theta_{\text{match}}, c_{\text{core}}, v_{\text{obl}}, g_{\text{guard}}$$

are high enough, but shell burden prevents admission or automatic high-priority status.

12C.2 RFOP GR/AC operator-like readout candidate

Target:

$$R_T = \text{RFOP GR/AC/continuum.}$$

Source:

$$R_j = \text{CST row 14: operator recovery.}$$

Coordinate	Score	Derivation
θ_{match}	H	both target an order/causal-region source to operator-like compatibility readout
c_{core}	H	operator-recovery core is structurally preserved
s_{shell}^-	H	CST operator shell must be replaced by RFOP region-system shell
κ_{cand}	H	candidate covers operator/compatibility face, not metric/dynamics/empirical faces
v_{obl}	H	obligations clear: define operator, prove stability, state target face
g_{guard}	VH	guardrails pass if non-dynamics boundary is retained

Status:

generated pending / high-priority within the operator-readout subtarget.

Not admitted because:

operator-like readout $\nRightarrow$ GR dynamics.

12C.3 Quantum measurement Schrödinger-dynamics extension

Target:

$$R_T = \text{BVD rows 13/35.}$$

Source:

$$R_j = \text{BVD row 12: Schrödinger dynamics.}$$

Coordinate	Score	Derivation
θ_{match}	M	dynamics face matches, but outcome/readout/update faces do not
c_{core}	H	unitary dynamics core is preserved
s_{shell}^-	H	measurement shell is largely missing
κ_{cand}	M	covers dynamics only
v_{obl}	M	obligations incomplete for outcome/readout
g_{guard}	VH	guardrails pass only as negative/insufficiency example

Status:

negative/weak.

This is a successful guardrail case: it shows why bounded Schrödinger dynamics does not solve measurement.

12C.4 Empirical likelihood/statistical calibrator

Target:

$$R_T = \text{BVD row 36.}$$

Source:

$$R_j = \text{empirical calibrator template.}$$

Coordinate	Score	Derivation
θ_{match}	H	target is formal-to-empirical bridge; source template matches calibrator need
c_{core}	H	likelihood/readout/admission core preserved
s_{shell}^-	M/H	target-specific data and instrument shell still required
κ_{cand}	H	covers empirical admission face
v_{obl}	VH	model, observable, likelihood, failure, update obligations explicit
g_{guard}	VH	guardrails pass if no empirical claim is made before data

Status:

high-priority empirical candidate.

Theorem 12C.5 — Score Provenance Theorem

A worked-case score is programme-derived when each coordinate can be justified by blocker graph preservation, transfer-core preservation, shell burden, candidate compatibility coverage, verification matrix clarity, and guardrail satisfaction.

Proof

These six items are exactly the coordinates of $\mathcal{G}_{\text{gen}}$, whose derivation is established by the Generative Score Pullback Theorem. Therefore a score that states its derivation in those six terms is programme-derived. $\square$

12D. Unblocker-selection similarity profiles

This section checks that the worked examples actually use the blocker-shape similarity machinery.

A source row should be selected only when its blocker profile has a nontrivial structural similarity profile:

$$\Sigma(\mathfrak{Bl}_j, \mathfrak{Bl}_T) = (\sigma_{\text{face}}, \sigma_{\text{transfer}}, \sigma_{\text{need}}, \sigma_{\text{cal}}, \sigma_{\text{dep}}).$$

12D.1 Hauptvermutung operator/spectral candidate

Target:

$$\mathfrak{Bl}_T = \mathfrak{Bl}_{\text{Haupt}}.$$

Source:

$$\mathfrak{Bl}_j = \text{operator/spectral recovery blocker}.$$

Similarity profile:

Coordinate	Value	Reason
σ_{face}	H	both use source structural data to constrain a target presentation
σ_{transfer}	H	both cross a source-data-to-target-structure gap
σ_{need}	H	both need an invariant/reconstruction operator
σ_{cal}	M/H	closeness calibrator must be target-specific
σ_{dep}	H	proof dependency has operator/invariant sufficiency form

Selection status:

eligible source candidate.

Not admitted because the source shell must be replaced by Hauptvermutung-specific similarity hypotheses.

12D.2 RFOP GR/AC operator-like readout candidate

Target:

$$\mathfrak{Bl}_T = \mathfrak{Bl}_{\text{GR}}.$$

Source:

$$\mathfrak{Bl}_j = \text{CST operator-recovery blocker.}$$

Similarity profile:

Coordinate	Value	Reason
σ_{face}	H	both involve causal/order/region source to operator-like readout both bridge source order/region data to continuum-relevant structure
σ_{transfer}	H	
σ_{need}	H	both need compatibility readout, not direct dynamics
σ_{cal}	M/H	RFOP target calibrator differs from CST operator setting
σ_{dep}	H	both require operator definition, stability, and target relevance proof

Selection status:

eligible high-priority candidate for operator-readout subtarget.

Boundary:

operator-readout $\nRightarrow$ metric dynamics.

12D.3 Quantum measurement Schrödinger-dynamics candidate

Target:

$$\mathfrak{Bl}_T = \mathfrak{Bl}_{\text{meas}}.$$

Source:

$$\mathfrak{Bl}_j = \text{Schrödinger dynamics bounded blocker.}$$

Similarity profile:

Coordinate	Value	Reason
σ_{face}	M	dynamics face overlaps, outcome/readout faces do not unitary evolution does not bridge to outcome/update
σ_{transfer}	M	

Coordinate	Value	Reason
σ_{need}	L/M	target needs outcome/readout, source supplies dynamics
σ_{cal}	M	formal dynamics calibrator differs from measurement/readout
σ_{dep}	M	proof dependency does not cover observer/outcome branch

Selection status:

negative-control / weak candidate.

This justifies why the paper treats Schrödinger dynamics as an insufficiency marker rather than an unblocker.

12D.4 Empirical calibrator candidate

Target:

$\mathfrak{Bl}_T$ = formal-to-empirical bridge blocker.

Source:

$\mathfrak{Bl}_j$ = empirical-calibrator packet.

Similarity profile:

Coordinate	Value	Reason
σ_{face}	H	formal model/readout/data faces align
σ_{transfer}	H	both bridge formal structure to observable data
σ_{need}	VH	both need model-to-observable calibrator
σ_{cal}	VH	calibrator role is direct
σ_{dep}	H	model, observable, likelihood, admission dependencies align

Selection status:

eligible empirical-calibrator source candidate.

Theorem 12D.5 — Worked-Case Unblocker Selection Audit Theorem

The worked examples use the unblocker-selection machinery correctly when each selected source row is accompanied by a structural similarity profile and source-status boundary.

Proof

Unblocker selection is governed by Sim_{Bl} , typed-match morphism, transfer-core/shell split, compatibility coverage, verification obligations, and guardrail status. The similarity profiles in this section expose the first of these fields for each worked source selection. The source-status boundary in §10G.1A prevents external theorem families from being treated as admitted packets without source cards. Therefore the worked examples now use the selection machinery explicitly rather than implicitly. $\square$

13. Reproducibility of generated examples

The worked examples are reproducible if a reader can identify:

1. the target row;
2. the source candidate rows;
3. the proposed typed match;
4. the transferable core;
5. the target-side shell conditions;
6. the verification obligations.

Theorem 13.1 — Seed-Table Generation Reproducibility Theorem

Given a diagnostic seed table whose rows include target regime, class, residual, active subtree, and action implication, a generative example is reproducible when it explicitly identifies the target row, source row, blocker profile, typed match, adaptation output, verification obligations, and guardrail result.

Proof

A generative example is a relation between a target blocker profile and a source unblocker packet. The seed table supplies the target/source profiles. The blocker profile supplies the search key. The typed match supplies the admissible relation. The adaptation output supplies the candidate. The verification obligations supply the admission test, the guardrail result states whether the candidate passes the required gates, and the generative decision vector supplies the status rule. Therefore another reader can reproduce or challenge the example by changing one of these declared components. $\square$

14. Explicit guardrails

The generative procedure is powerful only if it is guarded. Otherwise it collapses into analogy, metaphor, or source-context theorem laundering.

The following guardrails are not optional. They are part of the method.

Guardrail 14.1 — No analogy without typed blocker match

A candidate source problem may not be used merely because it sounds similar to the target problem.

The required condition is:

$$\Theta : \mathfrak{Bl}_j \rightarrow \mathfrak{Bl}_T$$

with preservation of blocker structure.

Thus:

$$\text{surface analogy} \not\Rightarrow \text{admissible generative match.}$$

Guardrail 14.2 — No import without adaptation

A source-context unblocker cannot be imported directly unless its shell conditions are also valid in the target context.

In general:

$$\text{Shell}_j \not\Rightarrow \text{Shell}_T.$$

Therefore the source unblocker must pass through:

$$\text{Adapt}_\Theta.$$

Guardrail 14.3 — No candidate without verification obligations

A generated candidate must include:

$$\Pi_T^* + \text{Obl}_T^*.$$

That is, the generated candidate must state what must be proved, constructed, calibrated, or empirically checked.

Thus:

$$\text{candidate generation} \not\Rightarrow \text{candidate admission}.$$

Guardrail 14.4 — No novelty without trace

A candidate may be novel, but it must not be untraceable.

Every generated candidate must preserve its genealogy:

$$\mathcal{U}_j \rightarrow \Theta \rightarrow \text{Adapt}_\Theta \rightarrow \mathcal{U}_T^*.$$

This is the programme's version of traceable novelty.

Guardrail 14.5 — No bypassing No-Free-Transfer

The generative method does not suspend No-Free-Transfer. It operationalizes it.

Every generated candidate must explicitly identify:

1. what structure transfers;
2. what structure does not transfer;
3. what target-side bridge laws are required;
4. what residuals remain.

Thus:

$$\text{source theorem} \not\Rightarrow \text{target theorem}.$$

Theorem 14.6 — Guarded Generation Theorem

A generated unblocker candidate is RFOP-admissible only if it satisfies all five guardrails:

$$G_1 \wedge G_2 \wedge G_3 \wedge G_4 \wedge G_5.$$

If any guardrail fails, the generated object is not an admitted candidate; it is either a loose analogy, an illegal import, an unverifiable proposal, an untraceable novelty claim, or a No-Free-Transfer violation.

Proof

A generative candidate is a relation-first transfer claim. By No Undeclared Relation, it must declare its source, bridge, target, and readout. By No-Free-Transfer, source-context validity does not transfer without declared bridge law. By Recursive Bridge Calibration, candidate admission requires proof or verification obligations under a calibrator. By Recursive Chart Learning and trace discipline, the generation path must be recorded. Therefore all five guardrails are required for admissible generation. $\square$

14.7 Guardrail failure table

Failed guardrail	Failure mode	Correct response
no typed blocker match no adaptation	loose analogy source-context theorem laundering	reject or refine blocker profile identify transfer core and redeclare target shell
no verification obligations	untestable suggestion	add proof/construction/calibration obligations
no trace	mystical novelty claim	record source, match, adaptation, and residuals
no NFT check	illegal transfer	identify target-type jumps and bridge laws

14A. Machinery-coverage audit

This section asks whether the paper fully excavates the programme machinery it depends on.

14A.1 Coverage table

Required machinery	Present?	Where used	Remaining gap
BVD question-shape	yes	Ω_T , seed rows, status/action	none at methodology level
blocker profile	yes	$\pi_{\text{Bl}}(\Omega_T)$	examples supplied; full external source-card population is application-level
closure-atlas search	yes	$\mathfrak{A}_U$, seed tables, $\mathfrak{A}_U^{\text{adm}}$ schema, canonical family population	exhaustive source-card population remains application-level
triadic substitution	yes	typed match Θ , blocker graph morphism	category-theoretic enrichment optional

Required machinery	Present?	Where used	Remaining gap
RCLT trace/update	yes	traceable novelty and candidate generation path	no full dynamic branch-field implementation yet
RBC calibrator	yes	$\mathcal{K}_T^*$, verification obligations, V_T	actual domain proof remains external
NUR/NUB/NUO/NFT	yes	five guardrails	no issue; guardrails explicit
BVD score-sheet generality worked examples	yes partial-to-yes	$\mathfrak{K}_{\text{gen}}^\Gamma$, $\Gamma_{\text{gen}}^{(5)}$ v0.5 matrices and v0.7 mediation	finer Γ_{count} optional examples generate candidates, not proofs
external/domain proof	no	outside methodology	must be done in target papers

14A.2 Gap classification

The remaining gaps are not foundation gaps. They are downstream application gaps.

Gap	Type	Does it block this paper?
source-admitted unblocker atlas	methodology schema and canonical family population now defined	not blocking; exhaustive source-card population application-level
typed-match model	blocker graph morphism now defined	category-theoretic enrichment optional
actual verification of a generated candidate	domain theorem/experiment	no; explicitly non-claimed and represented by V_T
empirical demonstration of yield	programme-validation	no for method paper; yes for future credibility
raw-count scoring refinement	granularity refinement	no; optional high-resolution layer

Theorem 14A.3 — Machinery Coverage Theorem

The paper fully covers and source-types the RFOP machinery required for a methodology-level theory of candidate generation, including typed matching, source-admitted unblocker atlas schema, and verification matrix framework, but it does not and cannot by itself supply the domain-specific verification of generated candidates.

Proof

A methodology-level theory of candidate generation requires: a way to diagnose the question, extract a blocker, search a branch atlas, match structural form, adapt a source unblocker, impose transfer guardrails, generate verification obligations, and record the trace. The paper supplies each of these and identifies its source in the programme stack. Domain-specific verification, however, requires theorems, constructions, data, or experiments inside the target field. RFOP methodology can declare those obligations but cannot satisfy them by method alone. Therefore the machinery is sufficient for candidate generation, but not for candidate admission. $\square$

Corollary 14A.4 — Full programme brought to bear at methodology level

At the methodology level, the full relevant RFOP programme is now brought to bear.

At the solution level, further branch-specific work remains required.

14A.5 Honest status

The paper now supports the claim:

RFOP has a programme-native theory of candidate generation.

It does not support the stronger claim:

RFOP automatically generates verified solutions.

14B. Methodology-level gap closure

v0.8 identified three remaining gaps:

1. source-admitted unblocker atlas;
2. fully formal typed-match model;
3. actual verification of generated candidates.

The typed-match model is closed in §5.4–§5.8.

This section closes the methodology-level versions of the other two gaps.

Definition 14B.1 — Source-admitted unblocker atlas

A **source-admitted unblocker atlas** is:

$$\mathfrak{A}_U^{\text{adm}} = (\{\mathfrak{U}_j\}_{j \in J}, \text{Src}, \text{Class}, \text{Scope}, \text{Verify}).$$

Where every $\mathfrak{U}_j$ satisfies:

1. it is backed by a source paper, theorem, construction, or verified domain result;
2. it states the blocker it resolved or reduced;
3. it states the source compatibility mechanism κ_j ;
4. it states the source calibrator $\mathcal{K}_j$;
5. it states the proof or verification structure Π_j ;
6. it states its non-transfer boundary Scope_j .

Definition 14B.2 — Atlas-admitted unblocker packet

An unblocker packet is atlas-admitted when:

$$\text{Adm}_{\mathfrak{A}_U}(\mathfrak{U}_j) = 1.$$

This requires source support and scope declaration.

If either is missing, the packet may be a candidate source, but it is not source-admitted.

Theorem 14B.3 — Source-Admitted Unblocker Atlas Existence Schema

Given any finite set of source-admitted solved or partially solved bridge problems with declared blocker profiles and compatibility mechanisms, there exists a finite source-admitted unblocker atlas.

Proof

For each source-admitted solved or partially solved bridge problem, form the packet $\mathcal{U}_j$ containing its source problem, blocker profile, compatibility mechanism, calibrator, proof/verification structure, and scope. The finite collection of such packets, together with source, class, scope, and verification maps, satisfies Definition 14B.1. $\square$

Register note

This theorem is an existence schema and the paper supplies a first canonical family population. It is not a claim that the atlas has already been populated exhaustively with source cards. A future application paper may build a complete concrete atlas.

Definition 14B.4 — Verification matrix

For a generated candidate:

$$\mathcal{U}_T^*,$$

define its verification matrix:

$$\mathcal{V}_T(\mathcal{U}_T^*) = (D_T, H_T, P_T, R_T, F_T, A_T).$$

Where:

Component	Meaning
D_T	target distinctions to be recovered
H_T	hypotheses / shell conditions required
P_T	proof or construction obligations
R_T	residuals expected to remain
F_T	failure conditions / falsifiers
A_T	admission criterion under $\mathcal{K}_T$

Definition 14B.5 — Verification status

A candidate has one of the following verification statuses:

Status	Meaning
unverified	obligations stated but not attempted
partially verified	some obligations discharged
blocked	a proof/falsifier blocks the candidate
residualized	candidate works only under bounded residual
admitted	all required obligations pass the target calibrator

Theorem 14B.6 — Verification Matrix Structural Completeness Theorem

A generated candidate is not eligible for admission unless its verification matrix declares target distinctions, hypotheses, proof obligations, residuals, failure conditions, and an admission criterion.

Here “complete” means **structurally complete as a required entry schema**. It does not mean that the verification procedure terminates in admission or refutation for every candidate.

Proof

Admission is a target-relative bridge claim. By No Undeclared Relation, the target distinction and bridge relation must be declared. By No Undeclared Boundary, hypotheses, residuals, and failure conditions must be declared. By No Undeclared Observer and RBC, the readout/admission criterion must be declared. Therefore all fields of V_T are necessary for candidate admission. This proves structural completeness of the matrix schema, not decidability or termination of the target-domain verification problem. $\square$

Corollary 14B.6A — Matrix completeness is not verification termination

complete verification matrix $\nRightarrow$ decidable verification outcome.

A candidate may have a structurally complete verification matrix and still remain unverified, residualized, blocked, or open in the target domain.

Corollary 14B.7 — Domain verification remains external

The methodology can require V_T , but it cannot fill P_T with a proof unless the proof is actually supplied in the target domain.

Thus the methodology-level gap is closed, while the domain-level obligation remains open.

14B.8 Closed methodology gaps

Former gap	v0.9 closure
source-admitted unblocker atlas	defined $\mathfrak{A}_U^{\text{adm}}$ and atlas-admission rule
typed-match model	defined blocker graph and typed-match morphism
verification matrix	defined V_T and completeness theorem
domain verification	explicitly marked external to methodology

14C. Derivation hardening audit

This section checks whether v0.10 has closed the “convenient declaration” risk.

14C.1 Machinery derivability table

Machinery	Derivable from programme?	Source / derivation
question-shape $\mathfrak{Q}_T$	yes	BVD diagnostic state
blocker profile $\mathfrak{B}l_T$	yes	BVD residual + theorem/no-go subtree + RBC obstruction vector
unblocker need Need_T	yes	BVD action space + residual repair target
blocker graph $\mathcal{G}(\mathfrak{B}l)$	yes	triadic face decomposition + target-type/NFT nodes + calibrator/readout
typed match Θ	yes	triadic substitution + charted readout preservation
match score θ_{match}	yes in form	BVD score-chart machinery applied to match preservation

Machinery	Derivable from programme?	Source / derivation
transfer core/shell split	yes	No-Free-Transfer; shell is the non-preserved context
adaptation operator	yes	RBC calibrated update of a bridge candidate
candidate-answer-shape	yes	GBS output of adapted candidates + obligations
guardrails	yes	NUR/NUB/NUO/NFT + RBC
verification matrix shape	yes	RBC + NUR/NUB/NUO admission conditions
status spectrum	yes in form	BVD granularity-indexed score chart
exact status thresholds	declared residue	chosen audit granularity, refineable
external source atlas content	declared/source-admitted residue	depends on actual external theorem inventory
empirical data/loss	declared residue	target-domain specific

14C.2 What cannot be derived without overclaim

RFOP cannot derive:

1. which external problem the user asks to analyze;
2. which tolerance a scientific field accepts;
3. which theorem inventory exists in a field;
4. whether a generated candidate is true in the target domain;
5. which empirical dataset or instrument will calibrate a prediction.

Trying to derive these would violate the programme's own boundary discipline.

Theorem 14C.3 — Derivation Completeness at Method Level

Generative Bridge Search is derivation-complete at the method level: all method-internal objects are derived from, or imported from, named RFOP programme machinery. Remaining declarations are application-specific residues and cannot be removed without overclaiming.

Proof

The method-internal objects are listed in §14C.1 and each has a programme source or derivation path. The remaining declarations listed in §14C.2 require external target choices, field tolerances, source theorem inventories, proof facts, or empirical data. These are not consequences of RFOP's formal machinery. If RFOP claimed to derive them, it would treat target-domain content as programme-internal without a bridge, violating No-Free-Transfer and No Undeclared Relation. Therefore the derivation is complete at the method level, while application residues remain explicitly declared. □

Corollary 14C.4 — No hidden convenience declaration remains

Any remaining declaration now has one of two statuses:

1. target/application residue;
2. audit-granularity choice.

No method-internal step remains declared merely because it is useful.

14C.5 Stronger honest status

The paper can now claim:

RFOP derives the generative method, not the truth of generated candidates.

This is the maximum safe claim.

14D. Cohesion and propriety audit

This section records the final targeted cohesion/propriety check.

14D.1 Flow check

The paper now follows a stable sequence:

source basis → programme derivation → core definitions → generation theorem → score/status machinery →

No later section is required to understand an earlier definition, except where a later audit explicitly validates an earlier construction.

14D.2 Propriety check

Check	Result
core machinery derived or imported from programme	pass
application residues explicitly bounded	pass
score/rubric not independent of BVD	pass
unblocker selection not analogy-based	pass
worked examples instantiate the machinery	pass after reconstruction and similarity cards
admitted solution not conflated with generated candidate	pass
external theorem families not treated as admitted packets without source cards	pass
remaining ambiguities localized	pass

14D.3 Remaining localized ambiguities

The only remaining ambiguities are legitimate application-level choices:

1. whether a target blocker graph has been drawn correctly;
2. whether a source theorem family has an adequate source card;
3. whether a typed-match morphism preserves enough structure;
4. whether shell burden has been counted fairly;
5. whether a target-domain verification succeeds.

These are not method-internal gaps.

Theorem 14D.4 — Cohesion/Propriety Theorem

The paper is cohesive and proper for its declared methodology target if every method-internal step is derived or imported, every remaining declaration is target/application-specific, every worked case exposes the reconstruction fields, and every generated candidate is separated from admitted solution.

Proof

The earlier sections derive the method-internal machinery. The declaration-residue audit restricts declarations to target/application choices. The reconstruction cards and similarity profiles expose the worked examples. The guardrail and verification sections prevent candidate generation from being read as solution admission. Therefore the paper is cohesive and proper for the declared methodology target. $\square$

15. Claims and non-claims

Claims

1. RFOP generation can be formalized as blocker-unblocker matching plus adaptation.
2. The method is programme-native: it derives from BVD, RCLT, RBC, triadic substitution, closure atlases, NUR, and NFT.
3. The method generates candidate unblockers, not guaranteed solutions.
4. Typed structural match is stricter than analogy.
5. Adaptation requires target-side shell conditions supplied by the application.
6. Verification remains substantive work.
7. Novelty can be traceable rather than mystical.
8. The method has five explicit guardrails: no analogy without typed blocker match; no import without adaptation; no candidate without verification obligations; no novelty without trace; no bypassing No-Free-Transfer.
9. A generated candidate is admissible only if it passes the Guarded Generation Theorem.
10. The broad BVD diagnostic table and deep CST diagnostic table can be used as generative seed sets.
11. A diagnostic row can function as a blocker profile, source blocker, candidate-unblocker source, or negative guardrail.
12. Worked generative examples are reproducible when target row, question-shape, blocker profile, source row, typed match, adaptation output, generative decision vector, verification matrix, guardrail result, and status are supplied.
13. The worked examples carry through end-to-end as generated candidate portfolios or negative-control cases, not as verified solutions.
14. The machinery is explicitly linked to the full RFOP programme dependency spine.
15. The generative score sheet is derived from the BVD score-sheet machinery and gives a calibrator form for match/adaptation decisions.
16. Candidate status is reproducible at ordinal $\Gamma_{\text{gen}}^{(5)}$ granularity, not left to expert vibes.
17. The candidate statuses form a coarse chart over a generative viability spectrum.
18. Finer generative granularities are admissible if supplied, monotone, and respectful of the admission gate.
19. A high-priority candidate is not an admitted solution; admission requires $\text{Adm}_{\mathcal{K}_T} = 1$.
20. BVD produces the question-shape of a bridge problem.
21. Generative Bridge Search produces the candidate-answer-shape.
22. The precise mediation is the blocker-profile projection followed by generative search:

$$M_{QA} = G_{\text{gen}} \circ \pi_{\text{Bl}}.$$

23. BVD and GBS are reciprocal faces of the same RFOP bridge-problem state.
24. The generative score vector is derived as a pullback/comparison readout between a BVD question-shape and a source unblocker packet along a typed-match morphism.
25. Reconstruction-card fields are programme-derived and not bookkeeping.
26. Generative Bridge Search is derivable from RFOP first principles as diagnostic readout, closure-atlas search, typed substitution, calibrated adaptation, guardrail admission, and trace/update.
27. The paper covers and source-types the RFOP machinery required for methodology-level candidate generation.
28. Every major component now has derivation/import status.
29. Typed structural match is formalized as a blocker-graph morphism.
30. A source-admitted unblocker atlas schema is defined.

31. A verification matrix framework is defined for generated candidates.
32. No method-internal component remains merely declared if it can be derived from the programme.
33. The only remaining declarations are target/application residues or audit-granularity choices.
34. Worked cases now have reconstruction cards and score provenance cards that instantiate the exact paper machinery.

Non-claims

1. This paper does not solve CST Hauptvermutung.
 2. This paper does not solve GR/AC/continuum recovery.
 3. This paper does not claim analogies are proofs.
 4. This paper does not license importing external theorems without adaptation.
 5. This paper does not claim RFOP automatically generates truth.
 6. This paper does not replace domain mathematics or empirical verification.
 7. This paper does not claim every structural match produces a useful candidate.
 8. This paper does not claim generated candidates are admitted solutions.
 9. The broad/deep diagnostic tables are seed sets for candidate generation, not proof of generated candidates.
 10. The end-to-end examples produce generated candidate portfolios, not completed mathematical or empirical closures.
 11. The generative score sheet ranks verification priority; it is not a truth score.
 12. The five-band status chart is not unique or final.
 13. Candidate-answer-shape is not the same as admitted answer.
 14. The paper defines a source-admitted unblocker atlas schema but does not yet populate a complete external atlas.
 15. The paper defines verification matrices but does not supply domain proofs for generated candidates.
 16. The paper does not claim application-specific targets, tolerances, datasets, or external theorem inventories are derivable from RFOP.
 17. The paper does not claim a complete verification matrix guarantees a terminating verification outcome.
 18. The first unblocker-atlas instance table is not an exhaustive atlas and does not replace source cards.
-

16. Next development

The methodology-level spine is now derivation-exhausted. The next developments are application-level.

1. populate a source-admitted unblocker atlas with actual theorem families such as Stone duality, Gelfand duality, spectral representation, Spec reconstruction, gluing theorems, Noether-type symmetry bridges, and empirical calibrator templates;
 2. run one fully source-admitted worked example starting from $\mathfrak{Q}_T$, computing $\pi_{\text{Bl}}(\mathfrak{Q}_T)$, selecting packets from $\mathfrak{A}_{\text{U}}^{\text{adm}}$, and producing V_T ;
 3. add a failure example where a plausible analogy is rejected by the blocker-graph morphism rule;
 4. add a raw-count Γ_{count} -style refinement for Γ_{gen} on the highest-stakes candidates;
 5. verify or reject one generated candidate in a target-domain paper;
 6. perform an independent row-by-row referee check of the reconstruction cards and decision-vector scores;
 7. expand the source-admitted external unblocker atlas with source cards;
 8. write the companion external application paper applying GBS to a foreign foundational programme such as loop quantum gravity, AdS/CFT, or asymptotic safety.
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17. Final statement

Bridge-Valence Diagnostics tells the programme what kind of problem it faces.

Generative Bridge Search tells the programme how to turn a diagnosed question-shape into a candidate-answer-shape.

The core mediated transition is:

BVD question-shape → blocker profile → typed structural match → candidate-answer-shape → verification ob
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v0.6 makes that beginning programme-linked, example-complete, decision-replicable, and granularity-aware at the methodological level: it generates candidate portfolios, scores their match/adaptation status under a BVD-derived calibrator, treats candidate status as a refineable spectrum, and states verification obligations without pretending that generation is proof.

End draft v0.16

AI-assistance disclosure

This manuscript was developed with AI-assisted drafting and review support. The author directed the research programme, selected and revised the mathematical and methodological content, and is responsible for all claims and errors.